

R2-24

Design and assembly manual

Two whole body prints • One print per side leg • 24 inches tall



September 12, 2026 | Revision C | Detailed exterior and metal moving frame

This is a digital prototype. Physical fit, frame strength, loaded driving, temperature, stopping distance and runtime require the commissioning tests. The video is a CAD simulation.

Package and reading order

10 STL designs; 14 installed printed pieces; 13 cutting profiles; 18 PNG drawings; 16 original MP3s; 220 wiring connections; a 30-second CAD video.

Budget estimate: USD 1896.56, including hardware, consumables, machining/welding allowances and six 1 kg filament spools. Listed prices and allowances are distinguished in the BOM. Tax, shipping, tools and printer are excluded. No parts were purchased.

Assembled mass estimate: 8.95 kg against the 9 kg design limit. The initial 6 kg assumption was not met. There is little mass reserve; weigh the actual parts and do not add payload. Drive performance with the specified TT motors is not physically verified.

Read the mechanical, fabrication and electrical chapters first. Follow all 60 assembly steps. The metal frame and loaded rolling tests are gates before final finish. Use the exact print manifest and hardware worksheet, not older revision files.

[Watch the CAD video](#) • [Download this PDF](#)

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Shape, size and printed parts

Revision C is nominally 609.6 mm high, with a 259.25 mm main body diameter. It has two whole stackable body prints, one complete dome and one print per side leg. There are 10 STL designs and 14 installed printed pieces. Small PCB spacers, structural shafts, bearings and strong brackets are purchased or fabricated in metal. No body quarters or leg segments are needed. The 13 DXF/SVG profiles are metal or insulating-sheet cutting files, not additional STLs.

The exterior follows the reference ellipse, radar-eye and body-skin drawings listed in appearance-research.md. The dome has an elliptical rise, six crown panels, a top disk, angular eye housing, convex lens, three projector cups, PSI disks and front/rear logic-display relief. The body includes utility-arm shapes, louvers, doors, coin slots, octagonal ports, couplers and a ribbed skirt. The side legs include horseshoes, hubs, boosters, struts, ankle bracelets and cylinders. These details are actual mesh relief and paint guides. They are not functioning display lamps, utility arms or projectors.

This is a scaled, adapted rear-foot robot. It is not a claim of exact current club-plan compliance. Feature placements without dimensioned references are image-derived. The rear-foot placement follows the user's wording; the familiar forward third-foot arrangement is not used here.

Coordinates are X across the shoulders, Y forward and Z up. Floor Z is 0. Side-foot centers are $X \pm 165, Y0$. Ground axes are local $Y \pm 37, Z31.5$. Wheel centers are nominally local $X \pm 29$. Check actual hub insertion on the purchased wheels; the stability calculation conservatively uses ± 26 . The side-foot exterior span is about 438 mm. Allow at least 500x600 mm clear floor area plus the turning radius. The rear-foot center moves from $Y = -128.926$ to -247.785 as posture changes. The front/back footprint also changes with body tilt and head rotation.

How the moving structure works

Two metal bearings support each 12 mm shoulder shaft. The aluminum leg spine clamps to that shaft through a split metal hub. Each side ankle is a rigid two-bolt metal joint. Extending the rear post therefore turns the body about the shoulder axis while the side legs remain supported on the floor. The legs swing relative to the torso. They are not independently powered walking limbs. All three feet stay down; there is no two-foot balancing mode.

The rear post has a separate square-tube sliding guide and a pin-ended P16 actuator. The guide carries sideways bending. The actuator changes length. A steel rod end and 12 mm transverse axle carry the rear-foot load. The steering servo operates a separate link and does not carry body weight. Its 8 degree yaw limit stays inside the rod end's 10 degree misalignment limit on a level floor. Pitch rotates about the axle rather than consuming that yaw allowance. Use gentle arcs; do not command a pivot turn.

Normal actuator stroke is 5.03832-72 mm. This produces approximately 0-15.054 degrees of body tilt. Both rear joint and side ankles keep their selected height. The fixed guide overlaps the inner tube by at least 160.5 mm across that range. Physical NC limits lie outside normal travel, near 2 and 80 mm. The software also checks position, current, progress, time and duty.

Load basis and acceptance limits

The design mass ceiling is 9 kg, including battery, finishes and fasteners. The initial 6 kg assumption was not met. The current estimate is about 8.95 kg, so there is little reserve and no payload allowance. This is a prototype commissioning limit, not a motor manufacturer's payload rating. Shaft calculations use 3x gravity, and actuator demand also includes a 1.25 allowance for alignment/friction. The calculation evaluates 101 positions and the stated center-of-gravity envelope. It finds 213.7 N maximum actuator demand versus the selected 300 N lifting rating, 93.7 MPa nominal shoulder-shaft equivalent stress and 68.8 MPa nominal inner-guide combined stress. These are screening calculations; they do not include every weld, surface defect, stress concentration or fatigue cycle. They do not certify the completed assembly.

Use shaft stock with a material certificate supporting at least 350 MPa yield strength, and 6061-T6 guide/frame stock. Do not substitute a soft threaded rod for a precision shoulder shaft. If quoting SKF6001 bearing ratings, fit the actual SKF bearings; the supplied housing bearings were not assigned those ratings. SKF SA12E data applies to that rod end, not to the surrounding clevis, welds or printed shell.

The loaded center of gravity used by the check is $X \pm 35$ mm, body-relative $Y = -60$ to $+20$ mm, and 80-170 mm below the shoulder axis. Weigh component groups and calculate their weighted coordinates. Verify horizontal balance with one scale under each foot. For level scales, $X_{cg} = 165 * (\text{right-left}) / \text{total}$ and $Y_{cg} = \text{rear}Y * \text{rear} / \text{total}$, with reactions in the same units. Check upright, mid travel and maximum tilt. Use the component mass/height calculation for vertical CG; three level scales alone do not measure CG height.

The requested TT motors remain the limiting drive components. Each has a plastic gearbox and press-fit output wheels. Two wheels on one motor share its torque. No continuous torque curve or shaft radial-load rating was supplied with the cited product data. Strong metal legs do not remove that limitation. Test the rolling chassis at the actual intended final mass on a dry, level, hard floor. Require reliable starts, no gearbox stall and no more than 0.5 A steady current per ground motor during gentle travel. Measure rear-foot rolling drag and keep it at or below 10 N, the calculation's acceptance ceiling. Reduce mass or revise the transmission if those tests fail; do not raise the current limits to force movement.

Ground PWM is capped at 90/255. Head PWM is capped at 70/255. These are duty commands, not measured speed. The rear-foot steering limit is 8 degrees. The smallest commanded turn radius is roughly 0.92m in the upright stance and increases as the rear foot extends. Tires still scrub in a tandem four-wheel foot. Test path tracking and stopping distance at low speed.

Body and head stack

The lower body mesh includes the skirt and starts at world Z130. Its main cylindrical wall starts at 165. The body seam is Z269.7. The upper body starts there and includes the fixed neck, ending at 454.3. The nominal dome datum is 456.8, leaving 2.5 mm fixed-neck clearance. The integrated raised crown reaches approximately 609.6 mm overall.

The lower seam has a 3 mm locating lip, with 0.3 mm radial clearance in the upper socket. Four M3x14 screws on radius 119 clamp the seam into captured nuts in the lower flange. The lower shell's four tabs sit on the metal base at Z173. Fit the upper body BEFORE the shoulder carriers, because its shoulder openings are closed circles rather than long service slots. Removing the upper body later requires releasing the shoulder assembly. Routine electronics access is through the removable dome and top plate.

The head floor is Z413-415. The printed bearing tower starts at 415 and holds 608 bearings at 415-422 and 434-441. A 12 mm metal spacer contacts their inner races. Nominally 0.8 mm of inner-race shims above the upper bearing places the integrated dome stem at 441.8. The cap surrounds this stem without touching it. The M8x60 bolt and captured nut connect the dome rotor through the inner-race stack. Set free rotation without axial play; do not squeeze the bearing seals with an oversized washer.

The dome underside includes its support spokes and a friction track at 65-88 mm radius. M7's wheel runs near radius 76.5. The holder bolts to the metal floor and has an open, dropped motor pocket. Its nominal shaft height is 426.3, giving approximately 1 mm tire compression against the track. Adjust with thin shims and the actual tire, aiming for 0.5-1 mm compression and reliable traction. Excess preload bends the shaft and increases current. Head weight is supported by the bearings, not M7.

Use the metal-fabrication worksheet for all shafts, plates, guides, clevises, secondary holes and fastener stacks. The CAD renders show purchased component envelopes; the worksheet defines machining details not visible in those simplified envelopes.

Print order and H2D settings

1. Obtain the actual motors, wheels, bearings and servo. Print one drive_cassette, bearing_tower, bearing_cap and head_motor_mount first. Check fits on the real components. These are installed parts, not extra disposable coupon designs.
2. Keep the native STL orientation. Body sections are base-down. The dome is supported under its integrated rotor/stem. The leg lies diagonally with its open inboard face toward the bed. Remove internal supports before inserting the spine. Print two identical leg copies; the symmetric leg is turned 180 degrees about world Z for the other side.
3. Print outer_foot twice: one as supplied and one mirrored in native X. Keep both toes forward. Rear_foot is a separate whole print. The three drive cassettes are identical. Do not mirror or rescale bearing fits.
4. Use the installed H2D 0.4 mm profile, 0.20 mm layers, four walls, five top/bottom layers, 20 percent gyroid, automatic normal supports including internal supports, and a 5 mm outer brim. Dome material is PLA; the remaining parts are PETG. The 1.2 mm body skin limits actual perimeter count locally. Never use vase mode. Inspect previewed thin relief.

5. All 10 current STL files sliced with those settings without a reported warning. The slice report includes exact STL hashes, predicted masses and times. Typical single-part predictions including discarded support are 519 g lower body, 608 g upper body, 883 g dome, 274 g leg and 258 g outer foot. These are software predictions, not physical print measurements.
6. Remove supports with the part fully cooled. Clear every bearing seat, lip, screw hole, motor slot and cable-tie slot. Ream a tight hole locally; do not scale an entire part to fix it. A bearing must seat by hand pressure without splitting its tower. Dry-fit the complete body seam.
7. Each STL fits within 300 mm per axis before brim. The checked H2D slices also account for the actual printer profile. Do not move a part into a nozzle exclusion area when rearranging a plate. No job was sent to a physical printer during this design run.
8. Weigh installed prints after removing supports. The solid-geometry plastic bound is recorded in cad/validation.json; it excludes purchased parts and supports. Final assembled mass and motor performance remain acceptance gates.

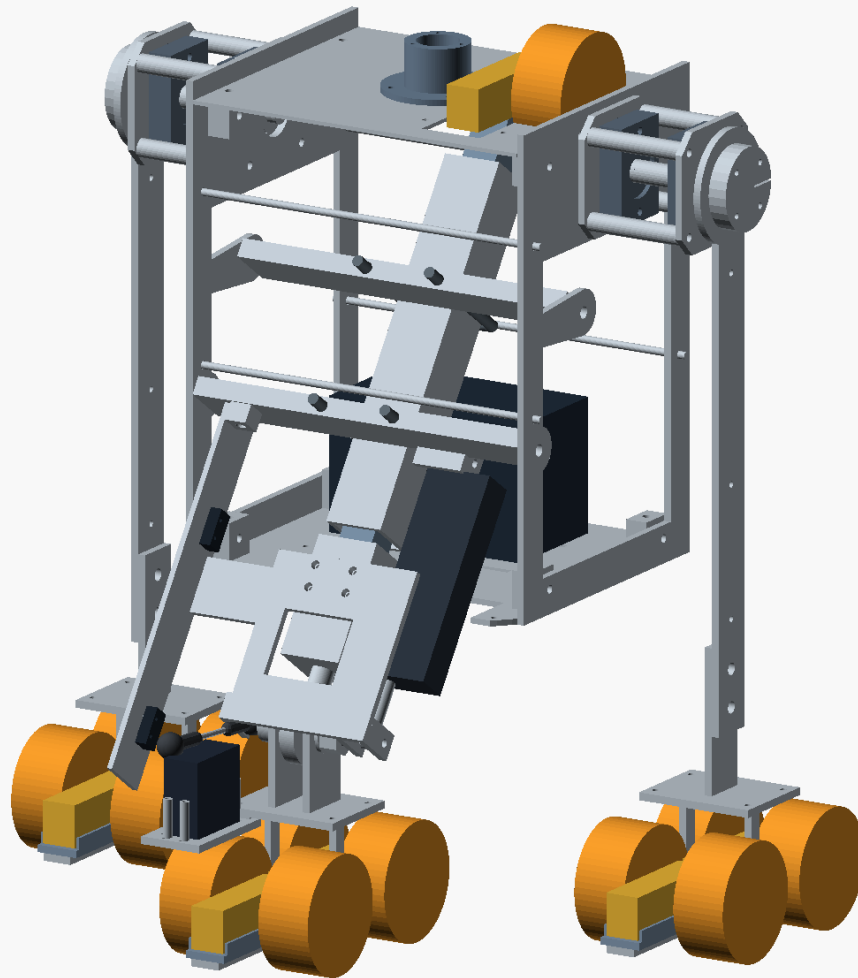
Finish and service

Paint the relief using the colored CAD sheets: white body and legs, silver dome and mechanical details, blue panel fields, dark recesses and the eye lens. There are six crown wedges, not eight generic stripes. Keep paint off bearings, threads, shaft clamps, wheel sockets, the body lip and the head friction track. Use thin coats rather than heavy filler.

Switch both supplies off and unplug the battery before service. Support the metal frame. Remove the dome's captured-nut connection while holding the lower bolt; then lift the dome. Remove the head-floor fasteners and unplug M7 to reach the electronics panels. Label each removable harness. Release panel ties only after supporting the panel. Removing a shoulder shaft or the rear actuator requires supporting the body independently; the post is part of its load path.

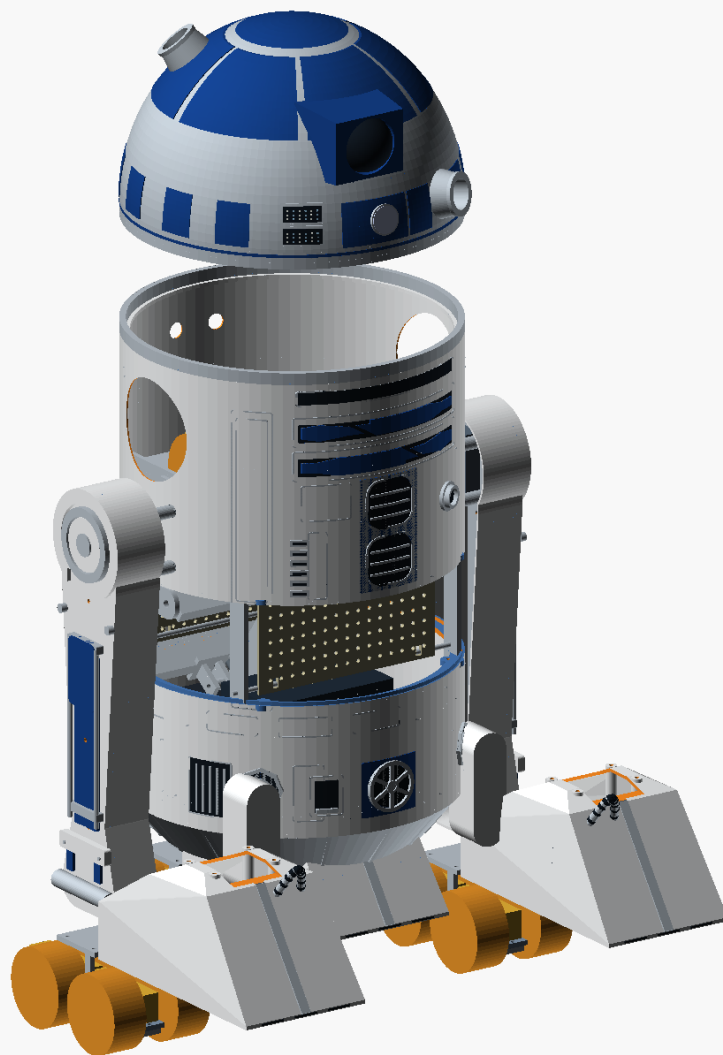
Before each use, inspect the wheel press fits, gearbox cases, shaft collars, clamp witness marks, clevis retention, guide pads, welds, wires and battery straps. Recheck after the first hour. Replace a worn gearbox or slipping wheel rather than forcing a cracked hub onto the shaft. Keep lubricant off tire treads, the head track and electronics.

Metal load frame



Metal shafts, spines, guide and foot frames carry body load. Electronics panels are omitted for this view; purchased hardware is shown as envelopes.

Whole stackable covers



Offsets identify the complete covers and are not assembly dimensions. The upper shell is installed before the shoulder carriers.

Rear post and foot



The rear post changes supported body posture. The rear foot has a separate steering link and a load-bearing spherical joint.

Metal fabrication worksheet

Drawing conventions and stock

All dimensions are millimetres. The 13 files in cad/metal are 1:1 DXF/SVG cut profiles. Set imported DXF units to mm. PNGs and PDF figures are identification drawings, not paper drilling templates. Profiles contain through-cut outlines and pilot/clearance holes. This worksheet adds thickness, depth, thread, fit and weld requirements. Do not infer a thread from the diameter of a rendered cylinder. Purchased parts in the 3D views are dimensional envelopes.

Use the thicknesses and quantities in bom/hardware.csv and the metal manifest. Flat frames, carriers, spines, crossbars and guide parts use 6061-T6 aluminum. Foot load frames and the moving yoke use weldable steel with a documented minimum 250MPa yield. Ground shoulder shafts require at least 350MPa yield. Insulating electronics panels use 3 mm G10 sheet. Do not mix steel and aluminum in a welded joint; use the bolted interfaces.

General cut dimensions: aim for ± 0.2 mm; mating hole centers ± 0.1 mm. Critical bearing bores use the stated fit, not the general cut tolerance. Deburr to a small smooth edge without enlarging bearing/shaft seats. Keep paired carrier and fork holes coaxial. Final reaming after assembly is preferred to trying to correct misalignment with loose fasteners. Maintain at least 1 mm measured clearance to moving hardware and shells.

Shoulder carriers and leg spines

metal_chassis is a 160x255x4 side-plate profile, with the world Z170-425 datum retained in the drawing. Cut two identical plates. The window is interrupted by integral webs at both guide-crossbar mounts. Do not remove those webs: they carry the rear-post force into the chassis. Shoulder axis is Y0,Z390. The right plate occupies X95-99; mirror this location for the left. Four electronics support rods use Y ± 70 ,Z275/365.

metal_carrier is 64x64x4 with chamfered corners. Cut four. The center 26 mm opening clears the 22 mm collars. Two 5.5 mm housing fastener holes are Y0,Z ± 21 about the shaft. Four 4.5 mm carrier holes are Y ± 25 ,Z ± 22 . Right carrier faces are X101.5-105.5 and 158.5-162.5. Four 53 mm spacers join them. Four 2.5 mm spacers separate the inner carrier from the chassis. M4x75 screws, washers and nuts clamp that stack. Repeat on the left.

Use four MISUMI BACA6001ZZ housings, 52x36x13. The bearing centers are X112 and 152 on the right, a 40 mm span. Mount each housing separately with two M5x16 screws through its carrier. Do not run one screw through two independently tapped housings. If relying on the cited SKF bearing data, replace the housing bearings with actual SKF6001-2RSH12x28x8 units. Retain the proper housing rings and shoulders. A replacement bearing must seat squarely; do not clamp a loose bearing with adhesive.

Cut two 12h5 ground steel shafts 90 mm long. Right-hand shaft ends are X92 and 182. Its inner collar spans 97.5-105.5; outer collar 158.5-166.5. Collars rotate inside the 26 mm plate openings. There is 1.5 mm nominal clearance from the outer collar to the leg pad. Use split collars of the listed envelope; verify bolt heads also clear the plate openings.

metal_spine is a 4 mm plate with a 60 mm shoulder pad and 25 mm wide lower spine. Cut two. Right spine is X168-172. Its ankle ends at Z106, with two 6.5 mm holes at Z119 and 139. Six skin-retention holes across the two spines are 3.4 mm, at 45, 155, 225 mm below the shoulder. The lower skin screw is at Z165, above the rigid ankle. Use M3x25 skin screws, allowing for the raised blue face on the middle mount. The four pad holes are 4.5 mm on 32 mm PCD at 45 degree increments. Countersink these on the inboard face for flush M4 screws. Do not remove more material than the actual screw-head angle and height require.

Machine two split hubs, 44 mm OD, 10 mm thick, with a 12H7 clamp bore. Tap the four pad holes M4, matching the 32 mm PCD. A 1.2 mm radial saw slit runs from the bore to the outer edge at Y positive, Z0 in the hub's face. The M4 clamp bolt crosses this slit at Y17, along Z. Machine flat 8 mm spotfaces near Z ± 5 for its washer and nut, leaving the clamp ligaments intact. Use the M4x20 bolt and locking nut. Pad screws are M4x10. The right hub spans X172-182. Check the shaft slips in before tightening, then mark the shaft/hub with a witness line to reveal later movement.

Base, top floor and electronics panels

metal_base is 3 mm aluminum at Z170-173. metal_headfloor is 2 mm aluminum at Z413-415. Cut one each. Their four main mounting holes are X $\pm$ 89, Y $\pm$ 55. Eight 12x20x20 aluminum corner blocks occupy X83-95 on the right and -95 to -83 on the left, Y45-65 or -65 to -45, at Z173-193 and 393-413. Cross-drill 4.5 mm at each block center in X and Z. The side-plate holes are Z183/403. Use M4x25 for horizontal joints and M4x30 for vertical joints, with accessible nuts. Clamp the frame square before tightening.

The base's rear notch clears the guide/actuator. Battery strap slots are at X $\pm$ 35, Y-7/47. Four shell tabs on radius 119 align with the extra M3 base holes. Keep those screws accessible until the lower shell is installed. Add a thin insulating/padding layer under the battery; do not use foam as a structural spacer at the chassis bolts.

The head-floor center hole is 16 mm. Four 3.4 mm holes on radius 24 at 45 degree increments mount the bearing tower. The rectangular window X30-64, Y-65 to 53 accepts the dropped motor holder. A separate wheel opening is X59-91, Y $\pm$ 32. Holder holes are X23/69, Y-50/45, outside the tire. The 190x160 floor terminates at the chassis inner faces. The holder's wings sit on top of the floor. No broad foot or head deck from an older revision belongs in this assembly.

panel_front and panel_rear are 160x100x3 G10 sheets. Their 10 mm grids are general mounting locations; the rear has a 69 mm speaker opening. Four M4x206 support rods cross the chassis at Y $\pm$ 70, Z275/365. Nuts and washers secure each rod to the side plates. Tie each panel independently to its two rods through the large corner slots. Panel planes are at Y70-73 and -73 to -70, centered on Z320. Use redundant ties and strain relief so harness pull cannot move a panel.

Use 6 mm nylon standoffs under boards. Arrange the Feather and low-current audio/sensor boards on the front panel. Place drive converters/drivers on the rear panel around the speaker and on available front-panel space. Keep tall items near the panel center: the cylindrical shell reduces available depth near its corners. Keep outer component faces within Y $\pm$ 100 and X $\pm$ 80, with additional corner clearance to the actual shell. Match-drill board holes using the purchased board after checking both faces for tracks/components. The grid is not a claim that every factory PCB uses 10 mm mounting pitch. UBECs and inline fuse holders use retained straps/ties; do not cover their heat-producing surfaces.

Welded four-wheel foot frames

Each foot begins with metal_sole, a 4 mm steel T profile. Its narrow part is 16x151, at X $\pm$ 8, Y-96 to 55, Z12-16. The transverse 44x8 feature at Y0 is part of the same cut profile. Do not replace it with a wide plate under the wheels. The motor-cassette screws are at X0, Y-90, -23, 52. The paired motor-tie slots must remain open after coating.

Weld two 4x8x55 stanchions to each T sole: X=-21 to -17 and 17 to 21, Y-4 to 4, Z16-71. Weld metal_bridge on top at Z71-75, centered X/Y. It is 64x50x4, with cover holes X $\pm$ 27, Y $\pm$ 17. Its lower face clears the 63 mm tires. The stanchions occupy the 11 mm longitudinal gap between axle rows; maintain at least 1 mm clearance after wheel tolerances and welds. Use controlled welds without large beads in the tire gap. Check each frame flat and square before fitting plastic or motors.

For each outer foot, weld metal_ankle vertically at X=-1 to 3, Y $\pm$ 12.5, Z75-152. Its two holes are 44 and 64 mm above its lower edge. The leg spine sits against its outboard face and bolts with two M6x20 screws. The left assembly is reflected across X. The side ankles are rigid; do not leave these bolts loose to make another pivot.

For the rear foot, weld two metal_fork ears at X=-12 to -8 and 8 to 12, Y $\pm$ 15, Z75-128. Their 12.1 mm bores are at Z113. Finish the inside gap to 16 mm and keep both bores coaxial. Two 3 mm inner-race spacers plus the rod end's 10 mm inner ring fill this gap. The outside rod-end head is only 8.5 mm wide, leaving room for its limited yaw.

The rear axle is Norelem 07534-12X25: steel 12.9, 12 mm shoulder, 25 mm shoulder length, M10x16 threaded tail and 18x9 head. Place a 1.3 mm total head-side shim between head and the first ear. This makes the clamp stack 25.3 mm, slightly longer than the shoulder, so the nut clamps the inner race rather than bottoming against the shoulder end. Use the 2 mm M10 washer and locking nut outside the far ear. Keep at least two full threads beyond the nut and at least 1 mm clearance to the adjacent actuator ear through the full yaw range. Dress only unused thread if needed; never shorten the load-bearing shoulder. Check free pitch/yaw after clamping.

The rear servo platform is 34x60x4 steel at X=-47 to -13, Y-80 to -20, Z66.1-70.1. Cut its 21.7x42.7 body opening at X=-40.85 to -19.15, Y=-71 to -28.3. Its 15x5x5 step joins it to the bridge at X=-32 to -17, Y-25 to -20, Z70-75. Four 3.4 mm-wide fore/aft slots

are centered at $X=-35/-25, Y=-74.4/-24.9$. Their 4 mm travel allows setup against the actual servo flanges. Four 20 mm steel M3 standoffs support the servo ears. Use uniform shims to set the horn's top face at Z109.5, centered at $X=-30, Y=-60$. The link's ball-center plane is Z113, not the case top. Measure the actual ear/shim/washer grip, then trim the M3x20 ear screws to give 5 mm thread engagement without meeting the opposite standoff screw. The dimensions of the actual servo flange and horn must be checked; the CAD case is an envelope. Verify a 1 mm gap from the linkage to the cover opening and from the servo to all wheel tops.

Telescoping guide, adapter and actuator

Guide coordinates are X transverse, N across its sloping face and T along the post, positive downward/rearward. The origin in body coordinates is (0,-40,240). T slopes 35 degrees rearward from vertical. Thus $Y=-40+N*\cos 35-T*\sin 35$ and $Z=240-N*\sin 35-T*\cos 35$ before body tilt. The outer guide spans $T=-150$ to 50. The inner tube ends at $L-62.5$, where $L=150+\text{actuator stroke}$, and is 270 mm long.

Use 31.75square x3.175wall outer tube and 19.05square x3.175wall inner tube. Fit four acetal pads per station near $T=-90$ and 30. Both complete 20 mm pad stations remain engaged even at maximum extension. Start with

3.175 mm pad thickness, then fit to actual tube corner radii and straightness. Target smooth, low-play sliding without a hard spot over the entire working range. Retain pads mechanically with two M2x5 countersunk fasteners per pad, at each station center plus/minus 5 mm along T. Keep at least 0.5 mm tip clearance to the sliding face. These are short retaining fasteners that stop inside the pad, clear of the sliding tube. Check actual wall and pad thickness before drilling. Do not drive a screw into the inner tube from the stationary outer guide.

Two 190x30x10 crossbars lie at $T=-75$ and 0. Their centers are $N=-30.875$. This puts their front face at $N=-15.875$ against the tube. Blind-drill 5 mm at each X end, 15 mm deep, and tap at least 12 mm usable M6 thread, for the chassis screws. Do not drill one 190 mm-long axial hole. Add 6.5 mm U-bolt holes along N at $X+/-21, T0$ relative to each bar. M6 square U-bolts have 36 mm inside width and 75 mm legs. Their curved/back portion bears across the outer tube; nuts and washers sit behind the crossbar. Tighten evenly without distorting the guide until the inner tube still slides freely.

On the $T0$ bar only, mill a 4 mm-deep front-face pocket at $X24-52$ across the 10 mm axial height. This receives the fixed actuator bracket flush. From the pocket bottom, drill and tap M4 at $X32/44, T0$, to 22 mm depth. Use M4x25 screws through the 4 mm bracket and washers, leaving tip clearance.

Machine the 34x34x65 adapter with a 19.2 mm square socket 35 mm deep from its proximal face. The opposite face has an M12x1.75 tapped bore 24 mm deep. Use the appropriate tap-drill diameter and thread gauge; the CAD pilot is not a finished thread. Two 5.5 mm cross holes are 10 and 26 mm from the proximal face. They also pass through the moving tube. Put 12.7 mm long steel crush sleeves inside that tube before tightening the M5x45 bolts. The stationary outer guide is not drilled for these bolts.

Four M4 blind holes on the adapter's rear N face hold the yoke. Their X coordinates are $+/-11$, with axial positions 17.5 and 29.5 from the proximal face. Drill to 7 mm and finish usable threads to 6.5 mm with a bottoming tap. M4x10 through the 4 mm yoke engages 6 mm. Confirm that no screw projects into the square socket or M12 bore.

Fit the SKF SA12E so its eye is 54 mm from the thread tip, with 21.5 mm thread engagement and the 6 mm jam nut seated. The eye then lies at $T=L$. The nut and adapter must clear the fork top through all body positions. Hold the rod end while tightening the jam nut; do not use the spherical joint as a wrench reaction point.

The actuator eye line is X38,N0. The fixed eye is T3 and moving eye $T=L$, giving 147+stroke millimetres between eyes. Its black case lies outside the guide, not inside it. Mount the fixed aluminum clevis to the $T0$ crossbar with two M4 screws. It has 10 mm inside spacing. Center the 8 mm rear eye with 1 mm spacers. Center the 6 mm front eye with 2 mm spacers in the moving yoke clevis. Use plain 4 mm steel pins 24 mm long, with 1 mm crossholes 1.5 mm from each end, outer washers and positive retaining clips. Do not use threads as the running surface inside an actuator eye.

metal_yoke is 4 mm steel. Its plate is behind the actuator at $N=-21$ to -17, not through the motor case. Weld the two 4x24x14 actuator ears at $X29-33$ and 43-47, $N=-17$ to 7, $T=L-7$ to $L+7$. Their pin holes lie at $N0, T=L$. Weld the 3.2 mm steering-link ear at $X=-37.5$ to -33.5 in the same N/T range. Machine the yoke's four mounting holes and cam tab from its DXF. Bevel the two axial cam-entry edges lightly and remove sharp burrs.

Limit rail and rear steering link

metal_limit_rail is 3 mm aluminum, in the $X=-76$ to -73 plane. Its drawing axes are N and T, spanning $N=-45$ to -15 , $T=205$. It remains outside the rear-foot shell. Its 12x13x10 clamp block occupies $X=-73$ to -61 , $N=-45$ to -32 , $T=15$. A M5 screw along T at $X=-67$, $N=-38.5$ attaches the block to the crossbar. Two M3x6 screws through the rail engage short blind holes in the block at $N=-42/-35$, $T=10$. Keep those blind holes clear of the M5 through bore; do not drill them through the block.

The actual SS-01GL mounting bores run across its 6.4 mm thickness, along X in this assembly. They do not run along N. Switch bodies are 19.8 mm along T and 10.2 mm along N. Their nominal mounting-hole line is $N=-29.3$. For nominal cam contact at T77 or 185, the two mounting holes are at $T=\text{contact}-14.5$ and $\text{contact}-5$. The rail slots permit N adjustment to set lever depression. Use M2x14 screws, washers and locking nuts.

The 30 mm moving cam spans $T=L-75$ to $L-45$, and reaches $X=-72.5$. At stroke 2, its retract-side edge reaches T77. At stroke 80, its extend-side edge reaches 185. Its length keeps the appropriate switch actuated throughout the remaining travel to either physical endpoint. Set the actual trigger with a meter and detached actuator. Aim near 2/80 mm, with normal 5.04-72 mm travel fully clear. Adjust to a small measured overtravel within Omron's

1.2 mm allowance; do not force the lever against the switch body.

The rear link joins guide point $Q=(-30,0,113)$ to a 12 mm servo crank about $S=(-30,-60,113)$, expressed relative to the rear foot's center. Link center distance is 61.188 mm at neutral. The guide-end stud axis is X; the horn-end stud axis is Z. This lets pitch rotate about the first stud while the spherical ends accommodate yaw. Use two Igus KBRM-03-MH rod ends with stainless-steel inner sleeves. Their verified sleeve clamping width is 6.1 ± 0.2 mm, head diameter 13 mm, housing thickness 4.5 mm, eye-to-threaded-mouth distance 18.5 mm and minimum usable M3 thread depth 5 mm. Their 30 degree limit is one-sided misalignment. The neutral guide-end skew is 11.31 degrees; check the entire yaw sweep with washers fitted.

At the guide ear's $X=-33.5$ face, use nominally 0.45 mm shims plus half the actual sleeve width to put the ball center at $X=-30$. At the horn top face Z109.5, the same nominal shim puts the ball center at Z113. Adjust to the measured sleeve width. Use small washers and verify that they clamp the sleeve without obstructing the housing; sleeve OD was not verified from the catalog and must not be inferred from housing thickness.

Install the guide M3x20 button-head screw from the bearing side. Its head faces the rear axle; the locking nut and surplus thread face away from that axle. Install the horn M3x16 button-head screw from beneath the horn. Check actual case and full-sweep clearance before tightening.

Two 18.5 mm eye offsets leave 24.188 mm between the threaded mouths. Nominal rod length 34.188 mm gives 5 mm insertion each side; measure actual usable depth and avoid bottoming. Adjust the finished assembly to 61.188 mm centers and lock the two link jam nuts without exceeding the 0.5 Nm manufacturer thread-torque limit. The sleeve clamping limit is 4 Nm; do not use that larger value on the M3 female threads.

Igus lists 400 N long-term tensile load along the link, but only 50 N long-term transverse load. The estimated servo-stall link load is below 70 N along the link. Do not apply the tensile rating to an off-axis load. Proof-test the detached link to 100 N and check free articulation. Vertical body load still bypasses this linkage. Set foot yaw to 0 before fixing the horn, then verify ± 8 degrees with a protractor and firmware calibration.

Inspection before powered assembly

Measure the carrier span, shaft fits, fork spacing, pin retention, adapter engagement, guide overlap and moving clearances. Lay each foot on a flat surface before fitting motors. Correct welding distortion in metal; do not pull the structure straight with plastic covers. Inspect all welds and edge ligaments. Proof-load the metal chassis in a supported fixture before putting the robot's weight through the TT gearboxes. The assembly chapter specifies that staged test and the later motor test.

Primary dimensional sources: [MISUMI bearing units](#), [SKF SA12E](#), [P16 datasheet](#), [Omron switch dimensions](#), [Norelem shoulder screw07534](#).

[Igus KBRM-03-MH dimensions](#), [Igus load and angle directions](#).

Electrical design and wiring

Architecture and power

Revision C uses the original Adafruit HUZZAH32 3405. Six Adafruit 3777 motors drive twelve ground wheels. A seventh 3777 drives one internal 3766 wheel under the dome. The rear steering servo is Adafruit 1142. The rear post uses an Actonix P16-100-256-12-P with position feedback. No wire crosses the rotating dome joint. Its display details are painted relief.

B1 is the protected Bioenno BLF-1206A 12 V / 6 Ah LiFePO4 pack, with its matched

14.6 V charger. Use its factory discharge and charge connectors. F1 is a

7.5 A fuse within 100 mm of the mating discharge plug. S1 MAIN feeds the logic branch through F6/1 A. S2 RUN feeds six separately fused 2 A branches: P1 left drive, P2 right drive, P3 rear drive, P4 steering/buffer, P5 head, and P6 actuator. P1-P5 are separate Adafruit 1385, rated 5 V / 3 A UBECs. P6 is the Pololu S13V25F12/4984 regulated 12 V converter. Never join converter outputs.

U6 Adafruit 4739 supplies 5 V logic/audio, up to 1.2 A. The Feather and sensors use its logic branch. Do not power any motor or servo through the Feather. The Feather BAT/JST socket stays empty. It is not a 12 V battery connector. Join all ground returns at the battery-negative distribution point. Run motor current through 18/22 AWG copper, not small PCB signal traces.

D1-D3 each provide two separate nominal 1 A current-limited DRV8833 channels. Both channels on a foot share direction inputs, but each has its own motor. Do not parallel motors onto one channel. Do not bypass the factory 0.2 ohm sense resistors. D4 uses only channel A for the head, also with the factory limit. Ground its unused B inputs; leave B outputs open. A low GPIO13 disables all four DRV8833 boards. Their open-drain fault outputs join at GPIO36 with a 10k pull-up to 3.3 V.

For supply sizing, allow 1.2 A per nominal 1 A motor channel, including headroom rather than treating the nominal limit as exact. At 5.25 V, seven channels plus 2 A steering and 1.2 A logic give 60.9 W. With 85 percent assumed converter efficiency and 11.2 V pack voltage, that is 6.40 A input. The 7.5 A main fuse and 12 A-continuous pack exceed that budget. Actual branch peaks, wire temperature and servo current must be measured. A stalled motor is not a permitted continuous load. During post motion, ground drive and head power commands are disabled. The actuator's roughly 12 W branch, steering and logic therefore do not add to simultaneous ground-drive load.

The UBEC input range is 6-16 V, so the 14.6 V fully charged pack is suitable. Use the input capacitors and short leads. Check every 5 V output is 4.75-5.25 V before connecting loads. Verify P6 is near 12 V and never connect it to the TT motors. The 72Wh nominal pack gives an estimated 2.0-3.3 hours at 15-25 W average and 70 percent usable energy. This is an assumption, not a runtime test. Measure runtime on the completed robot.

GPIO and named nets

GPIO numbers refer to the ESP32, not physical header position.

- 14/32: left forward/reverse, D1 AIN1+BIN1/AIN2+BIN2.
- 15/33: right forward/reverse, D2 AIN1+BIN1/AIN2+BIN2.
- 27/12: rear forward/reverse, D3 AIN1+BIN1/AIN2+BIN2.
- 13: D1-D4 SLP. External 10k pull-down; low disables these motors.
- 36/A4: D1-D4 FLT, external 10k pull-up to 3.3 V.
- 25/A1: rear steering, through U7 AHCT125 channel 1.
- 26/A0 and 17/TX: D4 head AIN1/AIN2. Separate 10k pull-downs.
- 4/A5 and 16/RX: post extend/retract through U7 and independent NC limits.
- 21 and 22/SCL: configured I2C SDA/SCL. The default Feather SDA header

is GPIO23, which this design instead uses for audio DIN.

- 39/A3: RUN sense via 10k/15k from P4 output.
- 34/A2: pack sense via 100k/22k from MAIN.
- 18/19/23: I2S amplifier BCLK/LRC/DIN.

GPIO12 is a boot-configuration pin. Keep its pull-down and add no pull-up. GPIO34,36,39 are input-only. All GPIO inputs stay within 3.3 V. The 5 V buffer outputs connect only to the servo and DRV8871, never back to the ESP32.

The 220-row wiring.csv and 12 connection sheets plus one power overview specify every connection, including grounds. Each horizontal circuit line is one connection; identical net names join across pages. Wire labels are authoritative. Some signals are repeated on two functional sheets for context.

Rear steering and independent post end limits

U7 must be 74AHCT125 DIP14. Pin 14 is P4/5 V, pin 7 ground, with 100nF directly across them. Output-enable pins 1,4,10 go to ground. Unused enable pin 13 goes to 5 V; unused input pin 12 goes to ground; unused output pin 11 is open. Do not substitute 74HC125. AHCT accepts 3.3 V control inputs at its 5 V supply.

Steering uses GPIO25 to U7 pin 2; pin 3 through 220ohm to SV1 signal. Servo red goes to P4, brown/black to ground. Use a metal 25-spline horn with a 12 mm link radius and the supplied center screw. Calibrate the actual neutral and pulse-to-angle response. Rear yaw is limited to 8 degrees each way. The software converts this to roughly 20 degrees at the servo using the

61.188 mm linkage, rather than assuming horn angle equals foot angle.

Post extend uses GPIO4 to U7 pin 5, output pin 6 to LS_EXT COM, and LS_EXT NC to D5 IN1. Post retract uses GPIO16 to U7 pin 9, output pin 8 to LS_RET COM, and LS_RET NC to D5 IN2. Fit 4.7k from each D5 input to ground AFTER the switch. Fit 10k pull-downs at both U7 control inputs. Opening a limit or breaking its signal wire removes only that direction command. The other direction remains available. This hardware gate does not depend on the ESP32 reading a switch. Do not put a bypass jumper across a limit.

Use Omron SS-01GL gold-contact switches. They carry about 1.06mA at 5 V through the 4.7k load, not actuator current. This meets the datasheet's 5 V/1mA reference minimum for the gold-contact family. Set LS_RET to open near 2 mm actuator stroke and LS_EXT near 80 mm. Both are outside the 5.04-72 mm normal operating range and inside the actuator's 100 mm physical stroke. NO terminals remain insulated and unused. The separate 30 mm cam stays on each switch through the remaining end travel. Adjust the bracket and cam depth so the lever is actuated without bottoming its mechanical travel.

Verify each limit with a meter first. Then, with the actuator detached from the robot and supported, hold the switch open and request its blocked direction: the motor must not run. Request the other direction: it must run away. Repeat for the second switch. Do not swap motor polarity alone: the named EXT/RET limits must still block their actual directions.

Actuator driver, feedback and fault response

D5 is Adafruit 3190/DRV8871. Its manufacturer schematic identifies the factory current-setting resistor as R1/30k. Remove that resistor and fit

71.5k1 percent on the provided resistor pads. Do not parallel the new resistor with the old one. The TI formula is $I_{trip}=64/R_k$, giving 0.895 A nominal. Using the 59-69kV datasheet range and 1 percent resistor tolerance gives about 0.817-0.975 A. This limits the driver; it is not a force rating.

P6 VOUT feeds INA219 VIN+. INA219 VIN- feeds D5 VM and the 470uF25 V output capacitor. D5 OUT1 goes to actuator red/pin 3; OUT2 to black/pin 4. Neither motor lead is ground. Do not fit a single ordinary flyback diode across this reversible motor. The driver's bridge provides recirculation.

Actuator orange/pin 1 is potentiometer ground, yellow/pin 5 is 3.3 V, and purple/pin 2 is the wiper. Wiper goes through 1k to ADS1115 AIN0. Add 100nF and 470k from AIN0 to ground. The pull-down makes a disconnected wiper read near zero. Sensor supply and I2C logic are 3.3 V. ADS1115 ADDR is ground for 0x48; unused analog inputs are grounded. INA219 address straps

A0/A1 are ground for 0x40. Use the breakouts' installed I2C pull-ups, not 5 V pulls.

Firmware reads ADS1115 AIN0 continuously at 128samples/s with ± 4.096 V range. INA219 uses the actual 0.1ohm shunt, calibration 4096 and 100uA current LSB. It also reads back sensor configuration. Calibrate POST_ADC_ZERO and POST_ADC_FULL with the detached actuator at its measured endpoints; the defaults 0/26400 represent ideal 0/3.3 V and are not measured values. Move the detached actuator to approximately 5 mm before installation. A factory-retracted 0 mm position is deliberately outside the healthy installed envelope. Disconnect motor power while changing calibration.

A posture button is held to move and released to stop. Motion requires an armed command lease, healthy sensors/power, stopped drive/head and centered steering. The controller stops for missing feedback, a position outside 1.5-85 mm, current above 0.75 A for 200ms, no 0.15 mm progress in 750ms, or 30 seconds travel. Faults latch until a power restart after the cause is fixed. Static feedback failure is detected by the no-progress test while moving; this is not a redundant position measurement.

P16 duty is 20 percent. Every move reserves an off interval four times its on time. A new press during that cooldown is rejected; release and press again after cooldown. A changed target during motion stops the actuator and requires release. No post motion resumes after disarm or Wi-Fi loss. The 500ms command-age limit is evaluated after bounded I2C transactions. Loop timing and physical stopping distance must be measured on hardware. The 2 second task watchdog is a separate response to a hung control task.

Pack sensing, audio, USB and harness

MAIN through 100k to GPIO34, then 22k to ground, gives 2.633 V at 14.6 V pack voltage. Add 100nF at the junction. Calibrate PACK_CORRECTION against a meter. Motion cuts off below 11.2 V for one second and requires 12.0 V to re-arm. Do not bypass the pack BMS. P4 through 10k to GPIO39, then 15k to ground, gives 3 V RUN sense. Add 100nF there. Losing P4 power disarms motion.

U8 MAX98357 A VIN is 5V_LOGIC. Its speaker connects only between its two speaker terminals; neither is grounded. GAIN goes through 100k to 5 V. Retain initial firmware volume 0.22. Keep I2S wires short and away from motor wiring. Original MP3s are included in the controller filesystem.

J_USB is a removable link between U6 output and the Feather USB power pin. Remove this link and unplug the pack BEFORE using computer USB. Otherwise two 5 V sources can back-feed. Flash and read the private Wi-Fi password with the controller disconnected from robot power. Remove USB, restore J_USB, then reconnect the pack. Never copy that password into a public repository or wiring label. Phone control uses the local R2-24 AP.

Use 18AWG for the main distribution, 22AWG for motor/servo branches and 26AWG for signals. Twist motor pairs; place 100nF at each of the seven motor terminals. Fit ferrules, insulated plugs, polarity labels and strain relief. Leave a restrained service loop for 67 mm post travel, 15 degree body motion and 8 degree rear yaw. Verify it through the entire range by hand before powering. Keep conductors clear of wheels, guide edges and shoulder shafts.

RUN off physically removes all motor and servo branch input power. MAIN off also removes logic. These are manual disconnects, not brakes. For charging, turn both off, unplug and remove the pack, then use its factory charge jack and matched charger. Reinstall with padding and two independent 20 mm straps. Do not open or modify the battery pack.

Primary references

[DRV8833 pin guide](#), [HUZZAH32 pin guide](#), [UBEC](#), [logic regulator](#), [SN74AHCT125](#), [DRV8871 datasheet](#), [Adafruit driver schematic](#), [Omron SS switch data](#), [P16 datasheet](#), [12V converter](#). Prices and stock were checked or carried as explicit allowances in the BOM on September 11-12, 2026. No parts were purchased.

Assembly and commissioning

Tools and build order

Use a Bambu H2D with 0.4 mm nozzle, dry PETG/PLA, calipers, square, flat reference surface, protractor, metric drivers/spanners, torque tools, multimeter, soldering equipment, ferrules, heatshrink and a current-limited 12 V bench supply. Metal fabrication also needs the machining/welding capability specified in fabrication.md. A spring/luggage scale measures rolling drag. Three platform scales help check balance. Use a temperature probe for the loaded motor test.

Build and test the metal structure before committing to finish work. The STL covers are not a substitute for the metal spines, shafts or foot frames. Keep a build log with actual part masses, measured clearances, calibration values, current, temperature and each acceptance result.

Stage 1 - inspect and fabricate

1. Read the mechanical, fabrication and electrical chapters together. Inventory the actual motors, wheels, bearings, P16 and servo against the BOM. Check stock and dimensions before placing any order. All prices in the hardware list are allowances, not a fabrication quote.
2. Open the main CAD and the 18 PNG drawings. Mark forward Y on all three foot assemblies. Side-foot centers are $X \pm 165$; rear placement is behind the body. The back foot must not be built as a forward foot.
3. Import the 13 cutting profiles in millimetres. Verify one known span with CAD measurement before cutting. Use the manifest's material, thickness and quantity. Apply the secondary operations in the fabrication worksheet rather than treating pilot holes as threads.
4. Make the two paired shoulder carriers, spines, guide crossbars, adapter and foot frames. Keep matching sets together. Check carrier hole alignment with a straight 12 mm reference shaft before coating.
5. Weld the steel T soles, stanchions, bridges and ankle/fork pieces in a square fixture. Keep weld beads out of the wheel envelopes. Check that both side uprights are square and the rear fork's gap is 16 mm. Correct metal distortion before installing plastic or bearings.
6. Finish the adapter socket, threads, crush sleeves, plain actuator pins and shaft hubs. Check all blind-hole depths and screw projection. Trial-fit the rod end to 21.5 mm engagement, then mark that position.
7. Dry-assemble the chassis base, side plates, corner blocks and head floor. Align all eight corner blocks before tightening. The head floor has separate motor and wheel clearances. Pass the actual head wheel through its intended swept space by hand.
8. Fit the two guide crossbars and square U-bolts. Install guide pads, then slide the inner tube by hand over the full intended range. There must be no hard spot or visible tube crushing. Retain each pad without any screw touching the sliding tube.

Stage 2 - first-fit prints and feet

9. Print one drive cassette, head bearing tower/cap and head motor holder using the recorded H2D settings. Clear their supports and holes. Seat the actual components by hand. Correct a tight hole locally rather than scaling the whole part.
10. Set a steel foot frame on blocks with its wheel locations accessible. Bolt the cassette to the T sole with three M3x14 screws, washers and nuts. Keep the cable-tie slots open through both metal and plastic.
11. Put both TT motor cans toward negative Y, with axles at $Y = -37/+37$. Add thin foam beneath each case. Fit two cable ties per motor, away from rotating shafts and electrical terminals. Leave the ties loose until axle height has been set.
12. Press one specified wheel onto each shaft end while supporting the opposite end. Do not hammer the gearbox. Keep equal hub engagement and clearance to the case. The drawing uses nominal $X \pm 29$ wheel centers; record the actual seating depth.
13. Place all four wheels on a flat reference surface. Adjust foam compression until both axles are at 31.5 mm. Tighten the ties without deforming the motor case. Turn every wheel and check the complete tire envelope, including the stanchion gap and

power-cell cover.

14. Repeat for the other two feet. Mark motors M1/M2 left, M3/M4 right, M5/M6 rear. Each motor will have its own driver output pair, even though two motors on one foot share a direction command.
15. Bolt each metal side spine to its ankle upright with two M6 screws. Check the datum holes are at Z119/139. Tighten these as rigid joints. The shoulder, not the ankle, is the side-leg pivot.
16. Fit the rear servo to its four steel standoffs. Use uniform shims and the platform slots to center the shaft at X=-30,Y=-60. Set the installed horn top face to Z109.5 and ball-center plane to Z113. Keep the horn off until electrical neutral is set. Check servo, standoffs and mounting screws clear the tires.

Stage 3 - shoulders and rear post

17. Fit the specified bearings in the metal shoulder housings. Bolt each housing to its carrier separately. Slide the ground shafts through the aligned bearing pairs; they must move without being forced.
18. Fit the inner/outer split collars in the stated locations. Put the split hubs and spine pads on the shafts. Check flush pad-screw heads, collar clearance and at least 1 mm from moving metal to each shell. Tighten clamps and apply witness marks across shaft/hub interfaces.
19. With the chassis supported independently, connect the inner guide tube to the adapter using its crush sleeves and two M5 cross bolts. Tighten against the sleeves, not the hollow tube walls.
20. Bolt on the moving yoke. Its plate sits behind the actuator's case. Fit the SA12E rod end and jam nut, preserving the marked engagement. Check both threaded and square sockets remain clear internally.
21. Put the spherical inner ring between the rear fork ears with its two 3 mm spacers. Install the specified shoulder axle bolt, 1.3 mm head shim, outer washer and locking nut. The shoulder must not bottom before the spacer/inner-ring stack is clamped.
22. Move the supported frame through the allowed pitch and yaw by hand. The spherical joint must turn freely without the bolt, nut or rod-end shank striking the yoke. Keep at least two exposed threads beyond the nut and 1 mm minimum measured clearance to the neighboring actuator ear.
23. Install the detached P16 between its fixed and moving clevises with the stated 1 mm/2 mm eye spacers, plain 4 mm pins, outer washers and four retaining clips. Its case is outside the sliding guide. Do not use the actuator rod to correct a crooked guide or bracket.
24. Install the stationary limit rail and two SS-01GL switches in their actual mounting orientation. Check the 30 mm cam passes their levers without hitting the switch cases or fasteners. Leave adjustment screws accessible for the bench calibration stage.
25. Do not install the rear steering link yet. The body remains supported while the post is calibrated. The factory-retracted actuator is not a valid powered operating posture in the installed robot.

Stage 4 - wire with power disconnected

26. Solder one 100nF capacitor directly across each of the seven motor terminals. Extend and twist the short motor leads, add strain relief, and fit separately labeled plugs. Insulate every exposed joint.
27. Build the main fuse and MAIN/RUN switch harness. Put F1 close to the discharge connector. Check the fuse values and wire gauge. Keep the factory battery connector intact and the pack unplugged.
28. Wire the five separate 5 V UBEC branches, logic converter and regulated 12 V actuator branch. Fit the listed capacitors with correct polarity. Use a common ground distribution point. Verify no two positive converter outputs have been joined.
29. Wire D1-D4 from the circuit schedule. Keep the factory DRV8833 current limits. Connect each ground motor to its own channel and M7 only to D4 channel A. Ground unused D4 B inputs and insulate unused outputs.
30. On D5 only, remove factory R1/30k and fit 71.5k1 percent. Inspect and measure the replacement before attaching the actuator. A parallel resistor would set the wrong current limit.

31. Build U7 on the solderable board. Check DIP orientation, decoupling, all output enables and the unused channel. Follow the post NC-limit paths exactly, with 4.7k pull-downs at D5 after the switches.
32. Wire ADS1115 and INA219 at 3.3 V logic, addresses 0x48/0x40. Check the P16 orange/yellow/purple feedback wires separately from red/black motor wires. Measure continuity to each named terminal; do not rely on color alone.
33. Wire pack/RUN dividers, the amplifier and speaker. Neither speaker wire is ground. Keep the Feather BAT/JST empty. Fit and label the removable J_USB power link. Check all GPIOs against config.h.
34. Mount boards on nylon standoffs and the insulating panels. Use actual board holes as drilling templates where the general grid does not align. Retain UBECS, fuses and speaker independently. Add service loops, but keep every conductor away from shafts, wheels and the guide.

Stage 5 - controller and detached actuator calibration

35. Unplug the pack and remove J_USB before connecting computer USB. Build with `pio run -d firmware`, then upload firmware and filesystem using `pio run -d firmware -t upload` and `pio run -d firmware -t uploadfs`. Use the actual controller's port if auto-detection is ambiguous.
36. Read the private 16-character AP password from the physical serial console at 115200 baud. Save it privately. Connect the phone to R2-24 and open <http://192.168.4.1>. Test the STOP button. Do not expect arming while required sensors or valid actuator position are absent.
37. Remove USB, restore J_USB and use a current-limited 12 V bench supply in place of the pack for initial rail checks. Keep all ground motors, M7 and the steering servo unplugged. Confirm 5 V rails are 4.75-5.25 V, P6 near 12 V and sensor logic near 3.3 V before connecting their loads.
38. Keep the actuator/guide assembly detached from the robot and secured on the bench. Record its factory-retracted eye spacing and measure voltage at ADS1115 AIN0. Compute the initial zero count as $\text{round}(V \cdot 8000)$. The firmware may latch an expected out-of-range fault at this position; do not bypass that installed-motion interlock.
39. For temporary bench control, unplug GPIO4/16 from U7 inputs 5/9. Leave their 10k pull-downs, both NC limits and all D5 pull-downs fitted. Use two normally-open momentary buttons from 3.3 V to U7 pin 5 or pin 9. Pressing one supplies extend or retract through the existing hardware limit. Release stops it. Insulate the disconnected ESP32 signal ends. Only the detached actuator is connected as a motor load for this test.
40. At low bench current, verify which button extends the rod. If wrong, correct the complete motor/direction/limit assignment before continuing. Do not merely relabel a reversed switch. Adjust LS_RET to open near 2 mm extension and LS_EXT near 80 mm, measured from the factory eye spacing.
41. Hold each limit open by hand and press its blocked-direction button. The motor must remain stopped. The opposite button must move away. Repeat for both switches. Verify the cam holds the switch open through the remaining mechanical travel, without bottoming the lever.
42. Stop at the measured 80 mm point and record AIN0 voltage/count. Set POST_ADC_ZERO to the recorded zero count and POST_ADC_FULL to $\text{ZERO} + (\text{COUNT}_{80} - \text{ZERO}) \cdot 100 / 80$. This two-point extrapolation avoids driving into the 100 mm mechanical endpoint. Check intermediate 10/40/70 mm positions with a ruler; require position error within 1 mm.
43. Return the detached actuator to approximately 5 mm. Remove the temporary buttons and restore GPIO4/16 wiring. Disconnect power and reflash the calibrated firmware with the USB isolation procedure. Restart at 5 mm and verify a healthy status before installing the mechanism under load.
44. Reconnect the steering servo alone, center it electrically, and install the metal horn with its supplied center screw. Set the 12 mm crank and 61.188 mm link length with the rear foot straight. Put the guide-end stud along X and horn-end stud along Z. Tighten both link jam nuts.
45. Check actual ± 8 degree rear yaw with a protractor. Adjust STEER_CENTER_US and STEER_US_PER_DEGREE to the measured servo, then rebuild if needed. Do not enlarge the yaw limit. Check link articulation and cover/harness clearance over all permitted body postures.

Stage 6 - supported structure and motion tests

46. Support the three metal foot bridges on a suitable fixture with all TT wheels unloaded. The rear support must be positioned for the selected post length. Check the bare metal structure first with a small load, then with the intended 9 kg total design load.
47. Have the fabricator check the load path and fixture before a staged static proof up to the 27 kg equivalent 3x-gravity load. Apply the load through the stated body CG region and keep it supported against a tip. Do not apply this proof load through the TT gearboxes. Inspect for cracks, slipped witness marks and residual deformation after unloading; any such change fails the test.
48. With normal load and independent support available, test short held posture commands. Release must stop movement. Check upright, middle and maximum commanded tilt. Verify guide overlap, rod-end clearance, stable support and the measured current. Do not increase the current thresholds to overcome a jam.
49. Measure on-time and cooldown. A two-second move must reserve at least eight seconds off before a fresh press can move again. A target change during motion must stop and require release. Verify the phone's position/tilt display against measured geometry; tilt is calculated, not read from a physical angle sensor.
50. Test STOP, button release, phone focus loss and Wi-Fi disconnect during a short supported move. Check no motion resumes after reconnection. Disconnect feedback during a supported test and confirm a latched stop. Restore it only with power off and correct the cause before restarting. Measure actual stopping delay and displacement.
51. Remove proof load and lower only the intended normal robot weight onto its wheels. Reconnect ground motors. Lift the wheels for first polarity tests. Each pair must propel its foot forward on a forward command; correct calibration::DIRECTION or matched motor connections as needed.
52. Test straight motion and gentle turns on a level hard floor at the lowest useful speed. Measure each ground motor's steady current and temperature. Require no stalls and no more than 0.5 A steady per motor. Stop for rapid heat rise, binding, tire rub or abnormal gear noise.
53. Measure rear-foot rolling drag with a spring scale and normal load. Require 10 N or less. Check the robot can start repeatedly at the intended final mass. Weigh all remaining covers and electronics before claiming the final load test is complete.

Stage 7 - final prints, head and finish

54. Print the remaining quantities in the manifest. Mirror only one outer foot in native X. Clean all supports and test each closed mesh's actual fastener access. Do not enlarge or cut away load-bearing metal to make a poorly cleaned plastic cover fit.
55. Fit the lower shell to the metal base and its captured seam nuts. Fit the complete upper shell before the shoulder carriers. Use the closed shoulder openings and the correct rear switch orientation. Tighten the four seam screws evenly, without crushing the locating lip.
56. Fit both complete leg skins over their metal spines and secure the six skin screws. Check the shoulder can swing without rubbing. Fit each whole foot cover to its four bridge bosses. Check screw tips, the power-cell housing, all tires and the rear linkage again.
57. Install the head bearing tower, both 608 bearings and 12 mm inner spacer. Add the nominal 0.8 mm upper inner-race shim stack and bearing cap. Fit the M8 bolt, lower inner-race washer and integrated dome rotor's captured nut. Adjust for free rotation without vertical play.
58. Fit M7 to its holder with foam and two ties. Fit its internal wheel. Bolt the holder to the metal floor, with the front screws beyond the wheel at Y45. Adjust tire compression to about 0.5-1 mm against the underside track. Turn the head by hand through a complete revolution before applying power; no metal plate or screw may enter the tire path.
59. Power M7 at low duty. Test both directions, release, STOP and lost Wi-Fi. Check drive traction and current without tightening the tire harder than needed. No wire may cross into the rotating dome. The decorative eye and display relief stay part of the single dome print.

60. Paint and label the covers, keeping fits and tire/track surfaces clean. Play all 16 original MP3s at low volume and check the speaker mount. Reweigh the finished robot, verify CG, repeat loaded driving/posture and stopping tests, and record runtime and temperatures. Keep the completed test log with the PDF. The design files alone do not prove that a built robot has passed these physical tests.

Regular checks

Before each session, check both physical switches, battery straps, wheel retention, shaft-clamp witness marks, pins, guide pads and harness loops. Stop and investigate any new play or noise. Disconnect and remove the pack for charging with its matched charger. Support the body before removing a shoulder or actuator. Keep the calibrated firmware and recorded mechanical measurements with the actual unit so replacement parts can be checked.

Original robot voice collection

The sixteen MP3 files are in `firmware/data/audio/`. They are original synthesized beeps, chirps and whistles. They are not recordings from a film, and they do not reproduce an actor's voice. Names describe intended moods.

`catalog.csv` provides names, durations and SHA-256 hashes. The phone sound menu plays every file through the onboard amplifier and speaker. The collection includes greetings, questions, agreement, concern, excitement, sleepy sounds, searching, surprise and a longer robot story.

Format: mono MP3, 22,050 Hz, 64 kbit/s. Rebuild with `python scripts/generate_audio.py`; NumPy and FFmpeg are required. The deterministic oscillator source and seeds are included. No web audio was downloaded. These generated audio files may be used, changed and shared with this robot project. No claim is made to the R2-D2 trademark.

After changing audio, run `pio run -d firmware -t buildfs`, then upload the filesystem to the disconnected controller as described in the manual. Firmware playback gain is limited to 0.22. Start with a low listening volume.

Digital verification and physical limits

This package contains a digital prototype. No physical robot was printed, assembled, driven or load-tested during this run. The assembly chapter's fit, strength, electrical and driving tests remain required.

Geometry and fabrication checks

The current print set is 10 STL designs and 14 physical prints. Mesh checks require a closed surface, consistent winding, positive volume, one connected solid, nonnegative bed Z and less than 300 mm extent on each axis. The actual H2D slices also check the installed printer profile rather than relying only on a nominal bounding box.

Fifteen CAD checks cover the body seam, shoulder openings, fixed neck, head tire/motor/floor, foot wheels and frame, both covers, rear steering envelopes and rear-post exit. Only explicitly named mating planes are allowed to touch. The body seam is checked from source with a 0.005 mm contact allowance: the installed CGAL version raised an assertion on re-imported coplanar STL surfaces. The source Boolean test resolves that numerical issue; exported STL connectivity and actual slicing are checked separately. These tests do not prove every physical part revision or fastener tolerance will fit.

The 22-row fastener worksheet checks nominal grip, washers, full nut engagement and blind-thread depth. Hex-pocket checks use across-flats dimensions: the 15.4 mm circumscribed dome pocket accommodates an M8 nut, and the 6.8 mm body pockets accommodate M3 nuts. The guide and horn hardware also require the measured rod-end sleeve width and full angular clearance.

The kinematic script evaluates 101 post positions. It keeps the rear pivot at 113 mm, retains both complete guide-pad stations and checks the support polygon over the stated CG envelope. Its 9 kg design case includes 3x gravity and the additional actuator alignment/friction allowance. Calculated demand is 213.7 N versus the actuator's 300 N lifting rating; nominal shaft stress is 93.7 MPa and inner-guide stress 68.8 MPa. These are section calculations, not a finite-element, fatigue or weld certification.

The initial 6 kg mass assumption was not met. The current budget uses cut-profile areas, material-density assumptions, G-code model extrusion and explicit purchased-component allowances. It is approximately 8.95 kg against a 9 kg limit, with little margin. Read `mass-budget.csv/json` for the current figures. Weigh the real parts; there is no payload allowance. This estimate does not establish that the selected TT motors can drive that mass on the builder's floor.

Actual slicer output

All ten parts are sliced with the installed H2D 0.4 mm profile, 0.20 mm layers, four walls, 20 percent gyroid, normal automatic supports and 5 mm brims. Each result records its current STL hash, material, full profile hash, return code, warning status, G-code hash and predicted material/time. Unchanged matching records can be reused; changed meshes or profiles are re-sliced. No printer connection or user-profile change is made.

Installed-model mass is separated from support/brim using labeled moving extrusion in the generated G-code. Machine purge/flush moves and stationary unretraction are excluded. This is a software estimate, not a weighing result. The current fourteen-copy model prediction is about 1.72 kg. Support material is additional filament that is discarded.

Firmware, wiring and sounds

PlatformIO builds the actual program and its 2 MiB filesystem for the original HUZZAH32. Both fit their partitions. Native C++ tests exercise arming, lease/replay rejection, command timeout, fault handling, bounds, reversal, post progress/current/feedback checks, cooldown and linkage geometry. The JavaScript test runs the actual phone code with a simulated page/network and checks release, cancel, focus loss, delayed replies and disconnect behavior. These are not radio-range or on-device tests.

The 220-row connection schedule is compared with firmware GPIOs, motor channels, head direction, independent NC actuator limits, feedback and I2C wiring. Divider upper bounds stay below 3.3 V at the stated voltages. These are static net checks, not continuity measurements on a harness.

All 16 original MP3s have matching SHA256 hashes and fully decode with FFmpeg. They contain 42.1 seconds of synthesized robot sounds. Physical ESP32 playback and speaker output remain part of commissioning.

Drawings, video and publication

The 18 PNGs include the assembled robot, orthographic and separated views, internal mechanisms, all ten printed components, thirteen cut profiles and the posture curves. CAD and image hashes identify the current revision.

The 30second1920x1080 video is a CAD simulation. It shows head rotation, rear-post motion, the exposed frame and separated covers. Its timing is illustrative. It is not footage of a built or tested robot. Source hashes, frame count, duration and full decode are checked before publication.

The repository's OIDC GitHub workflow uploads only the final PDF and MP4 to a dedicated private S3 bucket, served through CloudFront HTTPS. It downloads both anonymously, compares them with the committed bytes, checks MP4 range delivery and verifies direct anonymous S3 access stays denied. The final email is sent only after that workflow and PDF review.

Physical acceptance

The builder must verify purchased fits, head-wheel traction, actual steering/post positions, loaded mass/CG, joint retention, frame proof load, motor current, temperature, rolling drag, stopping distance and runtime. Keep the physical test log with the unit. Do not describe the robot as physically validated until those tests have passed.

Electrical bill of materials

USD prices are per item or stated pack. Research dates: September 11-12, 2026. Allowances are estimates. Follow exact part/rating notes and check current availability before ordering.

Qty	Component	Unit USD	Fit / rating / source
7	M1-M7 TT motor Adafruit 3777	\$2.95 (listed)	Six ground drive motors; one independent internal head friction motor Source
13	W1-W13 63mm TT wheel Adafruit 3766	\$1.50 (listed)	Twelve ground wheels; one internal head wheel; check stock and actual hub seating Source
1	U1 HUZZAH32 original Feather Adafruit 3405	\$19.95 (listed)	Original ESP32; not ESP32-S3 or V2 Source
4	D1-D4 Dual DRV8833 driver Adafruit 3297	\$5.95 (listed)	Leave factory1A limits intact; six ground channels and one head channel Source
5	P1-P5 5V3A UBEC Adafruit 1385	\$9.95 (listed)	Separate left/right/rear/steering/head5V rails; never parallel outputs Source
1	U6 5V1.2A logic converter Adafruit 4739	\$6.95 (listed)	Logic and audio only Source
1	SV1 MG-995 high-torque metal gear servo Adafruit 1142	\$19.95 (listed)	5V rear yaw;25 spline;40.7x19.7x42.9mm;62.41g; listed out of stock2026-09-12 Source
1	U7 74AHCT125 DIP14 Adafruit 1787	\$1.50 (allowance)	5V servo signal buffer; use exact AHCT type Source
1	U8 MAX98357A I2S amplifier Adafruit 3006	\$5.95 (listed)	Bridge output; neither speaker terminal is ground Source
1	SP1 3inch8ohm1W speaker Adafruit 1313	\$1.95 (listed)	77.8mm square envelope; foam and cable tie mounting Source
1	B1 Protected12V6AhLiFePO4 battery Bioenno BLF-1206A	\$79.99 (listed)	72Wh;12A continuous;0.7kg;112x66x71mm Source
1	CH1 Matched14.6V2A charger Bioenno BPC-1502DC	\$21.99 (allowance)	Remove battery for charging; use factory charge jack Source
2	S1-S2 20A30VDC SPST panel switch NKK S1A	\$12.00 (allowance)	MAIN and RUN;12.5mm panel holes; label OFF clearly Source
8	FH1-FH8 ATO inline fuse holder16AWG Littelfuse 0FHA0001ZXJ	\$4.00 (allowance)	Insulated holder; no exposed blades Source
1	F1 7.5A32V ATO fuse ATO7.5A	\$1.00 (allowance)	Main pack fuse
6	F2-F5 F7-F8 2A32V ATO fuse ATO2A	\$1.00 (allowance)	Separate inputs of five UBECs and actuator converter
1	F6 1A32V ATO fuse ATO1A	\$1.00 (allowance)	Logic input fuse
1	PP30 Anderson mating plug set PP30	\$6.00 (allowance)	Match factory discharge plug; do not cut pack lead Source
2	PROTO Half-size solderable board Adafruit 1609	\$4.50 (allowance)	Buffer and low-current distribution; motors use heavy wire Source
1	SOCKET DIP14 socket 14pin0.3inch	\$0.75 (allowance)	For U7
14	R10K 10kohm1percent0.25W resistor Metal film	\$0.10 (allowance)	Six drive inputs;enable;fault;steer;head2;post2;power sense
1	R15K 15kohm1percent0.25W resistor Metal film	\$0.10 (allowance)	Power sense bottom
2	R100K 100kohm1percent0.25W resistor Metal film	\$0.10 (allowance)	Pack sense top and amplifier gain
1	R22K 22kohm1percent0.25W resistor Metal film	\$0.10 (allowance)	Pack sense bottom
1	R220 220ohm0.25W resistor Metal film	\$0.10 (allowance)	Steering servo series resistor
11	C100N 100nF25V ceramic capacitor X7R	\$0.15 (allowance)	Seven motors;buffer bypass;two sense filters;actuator ADC filter
6	CI1-CI4 CI6-CI7 220uF25V electrolytic 105C low ESR	\$0.75 (allowance)	Five UBEC and actuator converter inputs
1	CI5 100uF25V electrolytic 105C low ESR	\$0.50 (allowance)	Logic converter input
5	CO1-CO5 470uF10V electrolytic 105C low ESR	\$0.75 (allowance)	Three ground drivers;logic;head rail

Qty	Component	Unit USD	Fit / rating / source
1	CS 1000uF10V electrolytic 105C low ESR	\$1.00 (allowance)	Servo rail
4	WIRE18 18AWG silicone wire metre Red black	\$1.50 (allowance)	Main power distribution
6	WIRE22 22AWG silicone wire metre Assorted	\$1.00 (allowance)	Motor and servo branches
5	WIRE26 26AWG stranded wire metre Assorted	\$0.60 (allowance)	Signal wiring
1	CONNECTORS Connectors ferrules headers heatshrink and labels Allowance	\$22.00 (allowance)	Six removable motor plugs;two servo extensions;USB power link;10A terminal blocks
1	USB Micro USB data cable USB-A to micro-B	\$5.00 (allowance)	For original HUZZAH32 flashing
1	ACT 100mm feedback linear actuator Actuonix P16-100-256-12-P	\$149 (allowance)	12V;300N;20percent duty;no internal limits;selected stroke5.04..72mm;backorder Source
1	P6 12V step-up/down regulator Pololu S13V25F12 / 4984	\$24.95 (allowance)	Actuator only;2A conservative branch design;22.9x22.9x9.7mm Source
1	D5 DRV8871 actuator driver Adafruit 3190	\$7.5 (allowance)	Replace factoryR1 30k with71.5k1percent;0.895A nominal limit;software0.75A/200ms Source
1	U9 16bit analog converter Adafruit ADS1115 / 1085	\$14.95 (allowance)	3.3V;0x48;AIN0 position;calibrate detached actuator ends Source
1	U10 High-side current sensor Adafruit INA219 / 904	\$9.95 (allowance)	3.3V logic;0x40;0.1ohm shunt;12V actuator supply branch Source
2	LS_EXT LS_RET Gold-contact hinge-lever limit switch Omron SS-01GL	\$3.5 (allowance)	NC contacts in separate5V driver-input paths;4.7k pulldowns;no motor current through switches Source
1	R_ILIM 71.5kohm1percent0.25W resistor Metal film	\$0.1 (allowance)	Replace D5 boardR1;never parallel with original
2	R_LIMIT 4.7kohm1percent0.25W resistor Metal film	\$0.1 (allowance)	D5 IN1/IN2 toGND after NC limits
1	R_POT 1kohm1percent0.25W resistor Metal film	\$0.1 (allowance)	Actuator ADC series filter
1	R_POT_FAIL 470kohm1percent0.25W resistor Metal film	\$0.1 (allowance)	ADC pulldown for disconnected wiper
1	CO6 470uF25V electrolytic 105C low ESR	\$0.9 (allowance)	Actuator12V rail;observe polarity
1	CAL_BUTTONS Normally-open momentary buttons;20pack Adafruit367	\$2.5 (listed)	Two used only for detached actuator commissioning; remove before operation Source

Mechanical hardware and consumables

Metal bolts, bearings and rods are purchased, not printed. STL mounts and adapters are in the print manifest. Quantities include stated spares and consumables.

Qty	Item	Specification	Use
4	metal_carrier	4mm 6061-T6; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
2	metal_chassis	4mm 6061-T6; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
2	metal_spine	4mm 6061-T6; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
1	metal_base	3mm 6061-T6; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
1	metal_headfloor	2mm 6061-T6; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
1	panel_front	3mm G10 insulating sheet; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
1	panel_rear	3mm G10 insulating sheet; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
1	metal_yoke	4mm steel; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
1	metal_limit_rail	3mm 6061-T6; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
2	metal_ankle	4mm steel; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
2	metal_fork	4mm steel; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
3	metal_sole	4mm steel; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
3	metal_bridge	4mm steel; exact DXF in cad/metal	Cut profile; secondary operations in fabrication.md
4	MISUMI bearing housing	BACA6001ZZ;52x36x13mm;12mm bore family	Shoulder carriers; retain original bearings as spares when fitting SKF replacements
4	Shoulder bearing	SKF6001-2RSH;12x28x8mm	Replace housing bearings if using the SKF load figures
2	Shoulder ground shaft	12h5 steel x90mm; certified yield>=350MPa	No printed shaft or threaded rod in shoulder bearings
2	Split shoulder hub	6061-T6;Machined44OD x10;12H7 bore;fourM4 on32PCD;M4 clamp	Couple spine to shaft; fabrication worksheet
4	Shaft collar	12mm split clamp;22OD x8 wide	Two per shoulder; no set screw against bearing surface
8	Carrier spacer	6061-T610OD x4.5ID x53mm	Four per shoulder between carrier plates
8	Inner carrier spacer	6061-T610OD x4.5ID x2.5mm	Core plate to inner carrier
2	Head608 bearing	608-2RS;8x22x7mm	Head bearing tower
1	Head inner spacer	Steel8.4ID x11.5OD x12mm	Between608 inner races
4	Head inner race shim	8.4ID x12OD;0.2mm each	Nominal0.8mm above upper head bearing; adjust to actual fit
1	Head spindle bolt	M8x60 button head steel;13.5mm head diameter	Rotating inner-race stack;clears16mm floor hole
1	Head M8 nut	M8 steel full nut;13mm AF	Captured in integral dome rotor
1	Head lower race washer	8.4ID x12OD x1.5mm steel	Under lower608 inner race
1	Outer guide tube	6061-T6;31.75square x3.175wall x200mm	Stationary post guide
1	Inner guide tube	6061-T6;19.05square x3.175wall x270mm	Moving post; do not drill stationary tube across its sliding path
8	Acetal guide pads	Nominal3.175thick x18wide x20long; fit to measured tubes	Four faces at each of two guide stations
2	Guide crossbar	6061-T6;190x30x10mm;blindM6 ends;two6.5mm U-bolt holes	Chassis stations t=-75 and0

Qty	Item	Specification	Use
2	Square U-bolt	M6 steel;36mm inside width;75mm legs;four nuts/washers total	Clamp outer tube against crossbars; tighten without tube distortion
1	Guide adapter block	6061-T6;34x34x65mm;19.2square socket;M12x1.75 tap	Moving tube to SKF rod end and yoke
2	Guide crush sleeve	Steel8OD x5.5ID x12.7mm	Inside moving tube at twoM5 cross bolts
1	Rear spherical rod end	SKFSA12E;12bore;M12 thread;10degree misalignment	Rear foot pitch and limited yaw; full shank engagement
1	M12 jam nut	M12x1.75;6mm thick	Lock rod-end adjustment with21.5mm thread engagement
1	Rear shoulder axle bolt	Norelem07534-12X25;12mm shoulder x25;M10x16 thread;18x9 head;steel12.9	Clamps spherical inner race through spacers;not a printed spindle
2	Rear axle race spacer	12.1ID x16OD x3mm steel	Center10mm inner race in16mm fork gap
6	Foot arch stanchion	Steel4x8x55mm	Two per foot;weld to T sole and bridge
1	Rear servo platform	Steel34x60x4mm;21.7x42.7body opening;four3.4mm slots	Weld atZ66.1;slots support adjustable servo posts
1	Rear servo step	Steel15x5x5mm	Join platform to rear foot bridge
4	Rear servo standoff	SteelM3 female-female;20mm;5OD;through-threaded	Adjust with shims to horn top planeZ109.5
40	Servo height shims	M3 steel;assorted0.5/1/2mm	Uniform0-10mm setup under servo ears;match horn plane
2	Moving yoke clevis ear	Steel4x24x14mm;4.5mm pin bore	Weld to yoke atX29 and43;10mm inside gap
1	Moving yoke steering ear	Steel4x24x14mm;3.2mm bore	Weld to yoke atX-37.5 to-33.5;shim to ball centerX-30
1	Fixed actuator clevis	6061-T6 bracket;28x24x4base;two4mm ears;10mm gap	Pins atX38,t3;twoM4 screws to t0 crossbar
4	Actuator eye spacers	M4ID;two1mm and two2mm;steel	Center8mm rear eye and6mm fore eye in10mm clevis gaps
1	Rail clamp block	6061-T6;12x13x10mm;M5 through;twoM3 blind	Attach stationary limit rail to t0 crossbar
1	Rear steering horn	Metal25-spline horn;12mm link radius;MG-995 compatible	Use correct spline and supplied center screw
2	Rear steering rod ends	IgusKBRM-03-MH;M3 female;3bore;B6.1+/-0.2;13mm head	Steel inner sleeve;400N catalog long-term along-link rating;no body-weight load
8	M4x75 bolts	Steel8.8;socket head	Shoulder carriers through all spacers
8	M5x16 bolts	Steel8.8;socket head	Bearing housings to carrier plates
12	M4x10 bolts	Steel8.8;8 countersunk and4 socket head	Spine-to-hub8;moving-yoke4
4	M4x20 bolts	Steel8.8;socket head	Hub clamps2;spares2
10	M4x25 bolts	Steel8.8;socket head	Horizontal chassis block joints8;fixed actuator bracket2
8	Chassis corner block	6061-T6;12x20x20;orthogonalM4 clearance bores	Base/head floor to side plates
4	M4 cross rod	SteelM4 x206mm;deburr ends	Carry front/rear electronics panels atY+/-70 Z275/365
4	M6x16 bolts	Steel8.8;socket head	Crossbar ends through chassis
4	M6x20 bolts	Steel8.8;socket head	Two rigid ankle joints per side
2	M5x45 bolts	Steel8.8;socket head	Adapter cross bolts through crush sleeves
1	M5x30 bolt	Steel8.8;socket head	Limit-rail clamp block to crossbar
2	Actuator clevis pin	4mm ground steel x24;1mm crossholes1.5mm from each end	Plain steel through both eyes;two retaining clips per pin

Qty	Item	Specification	Use
2	M3x6 bolts	Steel machine screw	Limit rail to blind-tapped clamp block
4	M2x14 switch bolts	Steel;washers;locking nuts	Two per limit switch;mount through actual6.4mm case thickness
13	M3x14 bolts	Steel machine screw	Body seam4;motor cassettes9
11	M3x12 bolts	Steel machine screw	Lower shell4;head tower4;spares3
12	M3x40 bolts	Steel machine screw	Four cover-to-metal-bridge screws per foot
4	M3x20 bolts	Steel machine screw	Head motor mount through floor;washer adjustment
4	M3x8 bolts	Steel machine screw	Servo standoffs to platform
3	M3x8 thread-forming screw	For2.6mm plastic pilots	Head bearing cap only
1	Guide steering stud screw	M3x20 button head;M3 locking nut;small washers	Head faces bearing;nut and excess thread face away from rear axle
60	M2.5 nylon standoff set	6mm female standoff;two bolts and nut each	Insulating electronics mounting;includes spare sets
4	M2 nut and washer set	Steel	Limit-switch mounting
80	M3 nuts and washers	Steel;include12 locking nuts;two washers per set	Body/foot/head mounts;spares included
40	M4 nuts and washers	Steel;include12 locking nuts;two washers per set	Carriers;panel rods;actuator pivots;spares
16	M5 nuts and washers	Steel;two washers per set	Bearing/guide/clamp joints;spares
12	M6 nuts and washers	Steel;two washers per set	U-bolts and ankles;spares
2	Battery straps	20x300mm hook-and-loop	Two independent straps through base slots
80	Cable ties	3mm x200mm nylon;minimum80N rated loop tensile	Two per motor;panel loops;speaker;harness;spares
1	Foam tape	1mm closed cell x1m	Motor setup and battery padding;do not cover regulators
1	Threadlocker	Removable medium strength;metal only	Shaft hubs and metal fasteners;not plastic or bearing races
1	Paint and masking tape	White silver blue black;thin compatible coats	Integral exterior relief;keep paint off fit surfaces
2	PLA filament	1kg silver or white;allowance	Dome slice uses883g including support;spare material
4	PETG filament	1kg white/gray;allowance	Other whole prints plus support and setup waste
1	Machining and welding allowance	Local shop quote required;no purchase made	Secondary machining;welded steel feet/yoke;surface finishing
1	Rear axle head shims	12.2ID x18OD;1.3mm total steel	Prevents shoulder from bottoming before inner-race clamping
1	Rear axle washer and nut	M10 DIN985 locking nut;10.5ID x20OD x2mm washer	Outer fork retention;at least two threads visible
4	Clevis-pin retaining clip	1mm wire R-clip;4mm pin compatible	Positive retention at both ends of each actuator pin
4	Clevis-pin outer washer	4.2ID x8OD x0.8mm steel	Outside actuator clevis ears
16	Guide-pad retaining screws	M2x5 countersunk;two per pad;check tip clearance	Retain acetal pads at T=-90/+30;minimum0.5mm from sliding face
6	M3x25 skin screws	Steel machine screw	13mm printed boss plus4mm spine;lower screw atZ165 clears ankle
8	M4x30 floor screws	Steel8.8;socket head	Base/headfloor through20mm corner blocks
1	Steering threaded link	M3 steel rod;40mm blank;cut after checking thread depth	Nominal34.188mm rod gives61.188mm centers at5mm insertion each

Qty	Item	Specification	Use
1	Steering thin nuts and shims	TwoM3 thin link jam nuts;3.2IDx6OD shims0.05/0.1/0.2mm;threadlocker	Nominal0.45mm sleeve-face offset;check free articulation
4	M3 servo ear screws	SteelM3x20;trim for5mm actual engagement	Measure ear+shim+washer grip;do not bottom against the other screw
1	Horn steering stud screw	M3x16 button head;M3 locking nut;small washer	From beneath horn through sleeve;check servo-case clearance

Mass budget

Estimated total 8952.5 g; limit 9000 g; estimated reserve 47.5 g. These figures combine calculation and allowances, not weighing. Use only installed quantities when measuring; unused stock and spares do not ride on the robot.

Component group	Estimated g	Basis
metal_carrier	140.6	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_chassis	448.3	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_spine	192.6	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_base	246.1	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_headfloor	130.7	Current cut-profile area x stock thickness x assumed density; all installed copies
panel_front	81.9	Current cut-profile area x stock thickness x assumed density; all installed copies
panel_rear	63.2	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_yoke	224.6	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_limit_rail	47.8	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_ankle	116.7	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_fork	92.6	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_sole	236.0	Current cut-profile area x stock thickness x assumed density; all installed copies
metal_bridge	298.0	Current cut-profile area x stock thickness x assumed density; all installed copies
All installed prints	1714.8	G-code labelled model extrusion; excludes supports/brim; not measured mass
Outer and inner guide tubes	343	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Two guide crossbars	308	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Moving adapter	162	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Eight corner blocks	104	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Six steel foot stanchions	83	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Rear servo platform and step	52	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Fixed actuator clevis	15	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Three moving yoke ears	32	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Limit-rail clamp block	4	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Two shoulder shafts	160	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Two aluminum split hubs	76	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Four aluminum bearing housings	178	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Four 6001 bearings	128	Component envelope/stock estimate or stated allowance; weigh actual installed parts

Component group	Estimated g	Basis
Two608 bearings	36	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Four steel shaft collars	68	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Aluminum carrier spacers	76	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Four electronics support rods	82	Component envelope/stock estimate or stated allowance; weigh actual installed parts
SKFSA12E	78	Component envelope/stock estimate or stated allowance; weigh actual installed parts
P16 actuator allowance	150	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Acetal guide pads	13	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Fasteners horn links shims and nylon standoffs	600	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Battery BLF-1206A	700	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Thirteen3766 wheels	494	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Seven3777 motors	214.2	Component envelope/stock estimate or stated allowance; weigh actual installed parts
MG-995 servo	62.4	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Electronics fuse holders connectors and installed wiring	600	Component envelope/stock estimate or stated allowance; weigh actual installed parts
Paint foam straps and ties	100	Component envelope/stock estimate or stated allowance; weigh actual installed parts

Fastener grip worksheet

Nominal stack checks complement the fabrication dimensions. Through joints include grip, washers and nut height. Tapped joints must have sufficient engagement and tip clearance. Measure actual hardware and never force a bottomed screw.

Joint	Fastener	Grip + washers	Nominal result
Body seam	M3x14	8 + 1 mm	2.60 mm beyond full nut
Shoulder carrier stack	M4x75	67.5 + 1.6 mm	2.70 mm beyond full nut
Chassis horizontal blocks	M4x25	16 + 1.6 mm	4.20 mm beyond full nut
Base through corner block	M4x30	23 + 1.6 mm	2.20 mm beyond full nut
Head floor through corner block	M4x30	22 + 1.6 mm	3.20 mm beyond full nut
Rigid side ankle	M6x20	8 + 3.2 mm	2.80 mm beyond full nut
Moving adapter cross bolt	M5x45	34 + 2 mm	4.00 mm beyond full nut
Head bearing tower	M3x12	5 + 1 mm	3.60 mm beyond full nut
Foot cover worst boss	M3x40	30 + 1 mm	6.60 mm beyond full nut
Motor cassette	M3x14	7.8 + 1 mm	2.80 mm beyond full nut
Leg skin with raised blue face	M3x25	20 + 1 mm	1.60 mm beyond full nut
Guide steering stud reversed head	M3x20 button	10.85 + 1 mm	4.15 mm beyond full nut
Horn steering stud	M3x16 button	8.85 + 0.5 mm	2.65 mm beyond full nut
Dome inner-race stack	M8x60 button	46.3 + 0 mm	7.20 mm beyond full nut
BACA housing	M5x16	4 + 1 mm	11.00 mm in 13 mm usable thread
Spine to split hub	M4x10 countersunk	4 + 0 mm	6.00 mm in 10 mm usable thread
Moving yoke to adapter	M4x10	4 + 0 mm	6.00 mm in 6.5 mm usable thread
Guide crossbar ends	M6x16	4 + 1.6 mm	10.40 mm in 12 mm usable thread
Fixed actuator bracket	M4x25	4 + 0.8 mm	20.20 mm in 22 mm usable thread
Limit rail short screws no washer	M3x6	3 + 0 mm	3.00 mm in 3.5 mm usable thread
Servo standoff base	M3x8	4 + 0.5 mm	3.50 mm in 20 mm usable thread
Acetal pad retention	M2x5 countersunk	3.175 + 0 mm	1.83 mm in 2.675 mm usable thread

Print manifest

Solid material bound 2079.0 g, before infill savings and without supports or purchased parts. Do not use this number as measured print or assembled mass. All dimensions below are millimetres. Use the native STL orientation and the reflection instructions in the mechanical chapter.

Part	Qty	Material	X x Y x Z	Solid g each
body_lower	1	PETG	259.76 x 267.25 x 142.7	253.91
body_upper	1	PETG	259.25 x 262.25 x 184.6	318.93
dome	1	PLA	259.55 x 282.24 x 167.8	519.31
leg	2	PETG	267.48 x 267.48 x 51.0	192.61
outer_foot	2	PETG	131.0 x 197.22 x 130.0	167.73
rear_foot	1	PETG	108.0 x 197.22 x 109.0	153.74
drive_cassette	3	PETG	22.6 x 151.0 x 9.7	18.95
head_motor_mount	1	PETG	58.0 x 110.0 x 12.9	25.26
bearing_tower	1	PETG	58.0 x 58.0 x 26.0	27.69
bearing_cap	1	PETG	34.0 x 34.0 x 3.0	2.59

H2D slice worksheet

Per-copy predictions from current G-code. Total filament includes discarded support and brim. Multiply by the quantity column for the complete set. These are not physical print measurements.

Part	Qty	Model g	Total filament g	Hours
body_lower	1	221.0	524.3	19.53
body_upper	1	276.3	609.1	20.72
dome	1	381.1	883.3	26.33
leg	2	147.6	273.9	7.04
outer_foot	2	157.9	254.5	7.85
rear_foot	1	144.5	236.1	6.87
drive_cassette	3	14.1	14.5	0.60
head_motor_mount	1	18.7	21.4	0.84
bearing_tower	1	17.6	18.5	0.83
bearing_cap	1	2.2	2.3	0.20

Point-to-point wiring schedule

One row is one connection. Identical named nets join across sheets. AWG identifies wire gauge. U7 output pin11 and unused D4 B outputs stay open. Read the complete electrical chapter before energizing.

Net	From	To	AWG
PACK+	B1 PP30 +	F1 7.5A input	18
FUSED+	F1 output	S1 MAIN input	18
MAIN+	S1 output	S2 RUN input	18
MAIN+	S1 output	F6 1A input	22
LOGIC_IN	F6 output	U6 MPM3610 VIN	22
RUN+	S2 output	F2 2A input	18
BUCK1_IN	F2 output	P1 UBEC IN red	22
BUCK1_IN	F2 output	C11 220uF25V +	22
GND	B1 PP30 -	P1 UBEC IN black	18
GND	B1 PP30 -	P1 UBEC OUT black	18
GND	B1 PP30 -	C11 -	22
RUN+	S2 output	F3 2A input	18
BUCK2_IN	F3 output	P2 UBEC IN red	22
BUCK2_IN	F3 output	C12 220uF25V +	22
GND	B1 PP30 -	P2 UBEC IN black	18
GND	B1 PP30 -	P2 UBEC OUT black	18
GND	B1 PP30 -	C12 -	22
RUN+	S2 output	F4 2A input	18
BUCK3_IN	F4 output	P3 UBEC IN red	22
BUCK3_IN	F4 output	C13 220uF25V +	22
GND	B1 PP30 -	P3 UBEC IN black	18
GND	B1 PP30 -	P3 UBEC OUT black	18
GND	B1 PP30 -	C13 -	22
RUN+	S2 output	F5 2A input	18
BUCK4_IN	F5 output	P4 UBEC IN red	22
BUCK4_IN	F5 output	C14 220uF25V +	22
GND	B1 PP30 -	P4 UBEC IN black	18
GND	B1 PP30 -	P4 UBEC OUT black	18
GND	B1 PP30 -	C14 -	22
5V_LEFT	P1 UBEC OUT red	D1 DRV8833 VMotor +	22
5V_LEFT	P1 UBEC OUT red	CO1 470uF10V +	22
GND	B1 PP30 -	D1 GND	22
GND	B1 PP30 -	CO1 -	22
LEFT_FWD	U1 GPIO14	D1 AIN1	26

Net	From	To	AWG
LEFT_FWD	U1 GPIO14	D1 BIN1	26
LEFT_FWD	U1 GPIO14	R_LEFT_FWD 10k pin1	26
GND	B1 PP30 -	R_LEFT_FWD pin2	26
LEFT_REV	U1 GPIO32	D1 AIN2	26
LEFT_REV	U1 GPIO32	D1 BIN2	26
LEFT_REV	U1 GPIO32	R_LEFT_REV 10k pin1	26
GND	B1 PP30 -	R_LEFT_REV pin2	26
ENABLE	U1 GPIO13	D1 SLP	26
FAULT	U1 GPIO36 / A4	D1 FLT	26
M1+	D1 AOUT1	M1 red	22
M1-	D1 AOUT2	M1 black	22
M1+	M1 red	CM1 100nF pin1	26
M1-	M1 black	CM1 pin2	26
M2+	D1 BOUT1	M2 red	22
M2-	D1 BOUT2	M2 black	22
M2+	M2 red	CM2 100nF pin1	26
M2-	M2 black	CM2 pin2	26
5V_RIGHT	P2 UBEC OUT red	D2 DRV8833 VMotor +	22
5V_RIGHT	P2 UBEC OUT red	CO2 470uF10V +	22
GND	B1 PP30 -	D2 GND	22
GND	B1 PP30 -	CO2 -	22
RIGHT_FWD	U1 GPIO15	D2 AIN1	26
RIGHT_FWD	U1 GPIO15	D2 BIN1	26
RIGHT_FWD	U1 GPIO15	R_RIGHT_FWD 10k pin1	26
GND	B1 PP30 -	R_RIGHT_FWD pin2	26
RIGHT_REV	U1 GPIO33	D2 AIN2	26
RIGHT_REV	U1 GPIO33	D2 BIN2	26
RIGHT_REV	U1 GPIO33	R_RIGHT_REV 10k pin1	26
GND	B1 PP30 -	R_RIGHT_REV pin2	26
ENABLE	U1 GPIO13	D2 SLP	26
FAULT	U1 GPIO36 / A4	D2 FLT	26
M3+	D2 AOUT1	M3 red	22
M3-	D2 AOUT2	M3 black	22
M3+	M3 red	CM3 100nF pin1	26
M3-	M3 black	CM3 pin2	26
M4+	D2 BOUT1	M4 red	22
M4-	D2 BOUT2	M4 black	22
M4+	M4 red	CM4 100nF pin1	26

Net	From	To	AWG
M4-	M4 black	CM4 pin2	26
5V_REAR	P3 UBEC OUT red	D3 DRV8833 VMotor +	22
5V_REAR	P3 UBEC OUT red	CO3 470uF10V +	22
GND	B1 PP30 -	D3 GND	22
GND	B1 PP30 -	CO3 -	22
REAR_FWD	U1 GPIO27	D3 AIN1	26
REAR_FWD	U1 GPIO27	D3 BIN1	26
REAR_FWD	U1 GPIO27	R_REAR_FWD 10k pin1	26
GND	B1 PP30 -	R_REAR_FWD pin2	26
REAR_REV	U1 GPIO12	D3 AIN2	26
REAR_REV	U1 GPIO12	D3 BIN2	26
REAR_REV	U1 GPIO12	R_REAR_REV 10k pin1	26
GND	B1 PP30 -	R_REAR_REV pin2	26
ENABLE	U1 GPIO13	D3 SLP	26
FAULT	U1 GPIO36 / A4	D3 FLT	26
M5+	D3 AOUT1	M5 red	22
M5-	D3 AOUT2	M5 black	22
M5+	M5 red	CM5 100nF pin1	26
M5-	M5 black	CM5 pin2	26
M6+	D3 BOUT1	M6 red	22
M6-	D3 BOUT2	M6 black	22
M6+	M6 red	CM6 100nF pin1	26
M6-	M6 black	CM6 pin2	26
ENABLE	U1 GPIO13	R_ENABLE 10k pin1	26
GND	B1 PP30 -	R_ENABLE pin2	26
FAULT	U1 GPIO36 / A4	R_FAULT 10k pin1	26
3V3	U1 3V	R_FAULT pin2	26
5V_SERVO	P4 UBEC OUT red	SV1 steering red	22
5V_SERVO	P4 UBEC OUT red	U7 AHCT125 pin14	26
5V_SERVO	P4 UBEC OUT red	CS 1000uF10V +	22
5V_SERVO	P4 UBEC OUT red	CB 100nF pin1	26
5V_SERVO	P4 UBEC OUT red	R_POWER_TOP 10k pin1	26
GND	B1 PP30 -	SV1 brown/black	22
GND	B1 PP30 -	U7 pin7	26
GND	B1 PP30 -	U7 pin1 / 1OE	26
GND	B1 PP30 -	U7 pin4 / 2OE	26
GND	B1 PP30 -	U7 pin10 / 3OE	26
GND	B1 PP30 -	U7 pin12 / 4A	26

Net	From	To	AWG
GND	B1 PP30 -	CS -	26
GND	B1 PP30 -	CB pin2	26
GND	B1 PP30 -	R_POWER_BOTTOM 15k pin2	26
5V_SERVO	P4 UBEC OUT red	U7 pin13 / 4OE	26
STEER	U1 GPIO25	U7 pin2	26
STEER	U1 GPIO25	R_STEER 10k pin1	26
GND	B1 PP30 -	R_STEER pin2	26
STEER_5V	U7 pin3	RS1 220ohm pin1	26
STEER_SIG	RS1 pin2	SV1 signal	26
RUN_SENSE	R_POWER_TOP pin2	U1 GPIO39 / A3	26
RUN_SENSE	R_POWER_TOP pin2	R_POWER_BOTTOM pin1	26
RUN_SENSE	R_POWER_TOP pin2	CP 100nF pin1	26
GND	B1 PP30 -	CP pin2	26
MAIN+	S1 output	R_PACK_TOP 100k pin1	26
PACK_ADC	R_PACK_TOP pin2	U1 GPIO34 / A2	26
PACK_ADC	R_PACK_TOP pin2	R_PACK_BOTTOM 22k pin1	26
PACK_ADC	R_PACK_TOP pin2	CA 100nF pin1	26
GND	B1 PP30 -	R_PACK_BOTTOM pin2	26
GND	B1 PP30 -	CA pin2	26
LOGIC_IN	F6 output	CI5 100uF25V +	22
GND	B1 PP30 -	CI5 -	22
GND	B1 PP30 -	U6 GND	22
5V_LOGIC	U6 5V OUT	J_USB removable link input	22
5V_LOGIC_USB	J_USB output	U1 USB pin	22
5V_LOGIC	U6 5V OUT	U8 MAX98357A VIN	22
5V_LOGIC	U6 5V OUT	CO4 470uF10V +	22
5V_LOGIC	U6 5V OUT	R_GAIN 100k pin1	22
GND	B1 PP30 -	U1 GND	22
GND	B1 PP30 -	U8 GND	22
GND	B1 PP30 -	CO4 -	22
I2S_BCLK	U1 GPIO18	U8 BCLK	26
I2S_LRC	U1 GPIO19	U8 LRC	26
I2S_DIN	U1 GPIO23	U8 DIN	26
GAIN	R_GAIN pin2	U8 GAIN	26
SPK+	U8 speaker +	SP1 +	22
SPK-	U8 speaker -	SP1 -	22
RUN+	S2 output	F7 2A input	18
HEAD_BUCK_IN	F7 output	P5 UBEC IN red	22

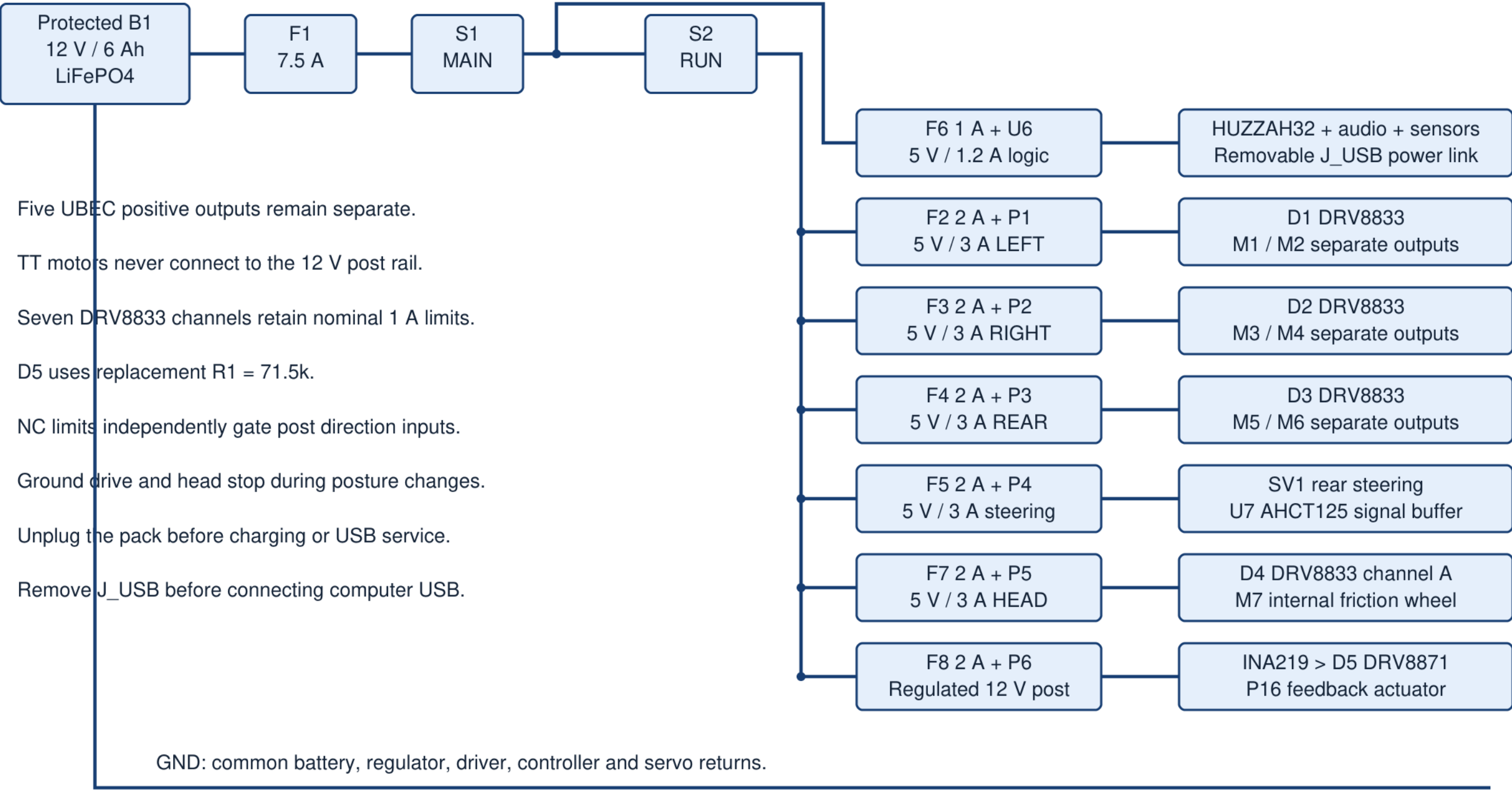
Net	From	To	AWG
HEAD_BUCK_IN	F7 output	CI6 220uF25V +	22
GND	B1 PP30 -	P5 IN black	22
GND	B1 PP30 -	P5 OUT black	22
GND	B1 PP30 -	CI6 -	22
GND	B1 PP30 -	CO5 -	22
GND	B1 PP30 -	D4 GND	22
GND	B1 PP30 -	D4 BIN1	22
GND	B1 PP30 -	D4 BIN2	22
5V_HEAD	P5 OUT red	D4 VMotor +	22
5V_HEAD	P5 OUT red	CO5 470uF10V +	22
ENABLE	U1 GPIO13	D4 SLP	26
FAULT	U1 GPIO36 / A4	D4 FLT	26
HEAD_FWD	U1 GPIO26	D4 AIN1	26
HEAD_FWD	U1 GPIO26	R_HEAD_FWD 10k pin1	26
GND	B1 PP30 -	R_HEAD_FWD pin2	26
HEAD_REV	U1 GPIO17	D4 AIN2	26
HEAD_REV	U1 GPIO17	R_HEAD_REV 10k pin1	26
GND	B1 PP30 -	R_HEAD_REV pin2	26
M7+	D4 AOUT1	M7 red	22
M7+	M7 red	CM7 100nF pin1	26
M7-	D4 AOUT2	M7 black	22
M7-	M7 black	CM7 100nF pin2	26
RUN+	S2 output	F8 2A input	18
POST_BUCK_IN	F8 output	P6 S13V25F12 VIN	22
POST_BUCK_IN	F8 output	CI7 220uF25V +	22
12V_POST_RAW	P6 VOUT	U10 INA219 VIN+	22
12V_POST	U10 VIN-	D5 DRV8871 VM	22
12V_POST	U10 VIN-	CO6 470uF25V +	22
GND	B1 PP30 -	P6 GND	22
GND	B1 PP30 -	CI7 -	22
GND	B1 PP30 -	CO6 -	22
GND	B1 PP30 -	D5 GND	22
GND	B1 PP30 -	U10 GND	26
GND	B1 PP30 -	U9 ADS1115 GND	26
GND	B1 PP30 -	U9 ADDR	26
GND	B1 PP30 -	U10 A0	26
GND	B1 PP30 -	U10 A1	26
GND	B1 PP30 -	U9 AIN1	26

Net	From	To	AWG
GND	B1 PP30 -	U9 AIN2	26
GND	B1 PP30 -	U9 AIN3	26
POST_EXTEND	U1 GPIO4	U7 pin5	26
POST_EXTEND	U1 GPIO4	R_POST_EXTEND 10k pin1	26
GND	B1 PP30 -	R_POST_EXTEND pin2	26
POST_EXTEND_5V	U7 pin6	LS_EXT COM	26
POST_EXTEND_LIMITED	LS_EXT NC	D5 IN1	26
POST_EXTEND_LIMITED	D5 IN1	R_LS_EXT 4.7k pin1	26
GND	B1 PP30 -	R_LS_EXT pin2	26
POST_RETRACT	U1 GPIO16	U7 pin9	26
POST_RETRACT	U1 GPIO16	R_POST_RETRACT 10k pin1	26
GND	B1 PP30 -	R_POST_RETRACT pin2	26
POST_RETRACT_5V	U7 pin8	LS_RET COM	26
POST_RETRACT_LIMITED	LS_RET NC	D5 IN2	26
POST_RETRACT_LIMITED	D5 IN2	R_LS_RET 4.7k pin1	26
GND	B1 PP30 -	R_LS_RET pin2	26
POST_RED	D5 OUT1	ACT red / pin3	22
POST_BLACK	D5 OUT2	ACT black / pin4	22
POST_ILIM	D5 ILIM / IC pin4	D5 replacement R1 71.5k pin2	26
GND	B1 PP30 -	D5 replacement R1 pin1	26
3V3	U1 3V	U9 VDD	26
3V3	U1 3V	U10 VCC	26
3V3	U1 3V	ACT yellow / pin5	26
GND	B1 PP30 -	ACT orange / pin1	26
POT_WIPER	ACT purple / pin2	R_POT 1k pin1	26
POST_ADC	R_POT pin2	U9 AIN0	26
POST_ADC	U9 AIN0	R_POT_FAIL 470k pin1	26
GND	B1 PP30 -	R_POT_FAIL pin2	26
POST_ADC	U9 AIN0	C_POT 100nF pin1	26
GND	B1 PP30 -	C_POT pin2	26
I2C_SDA	U1 GPIO21	U9 SDA	26
I2C_SDA	U1 GPIO21	U10 SDA	26
I2C_SCL	U1 GPIO22	U9 SCL	26
I2C_SCL	U1 GPIO22	U10 SCL	26

Circuit sheets

R2-24 | revision C power distribution

Architecture view. Follow wiring.csv and the connection sheets for exact pins, returns and capacitor polarity.



Fuses are on converter inputs. MAIN off removes all supply power. RUN off removes every motor and servo branch.

00-power-overview.svg | Named nets connect across sheets. Full return wiring is in the schedule.

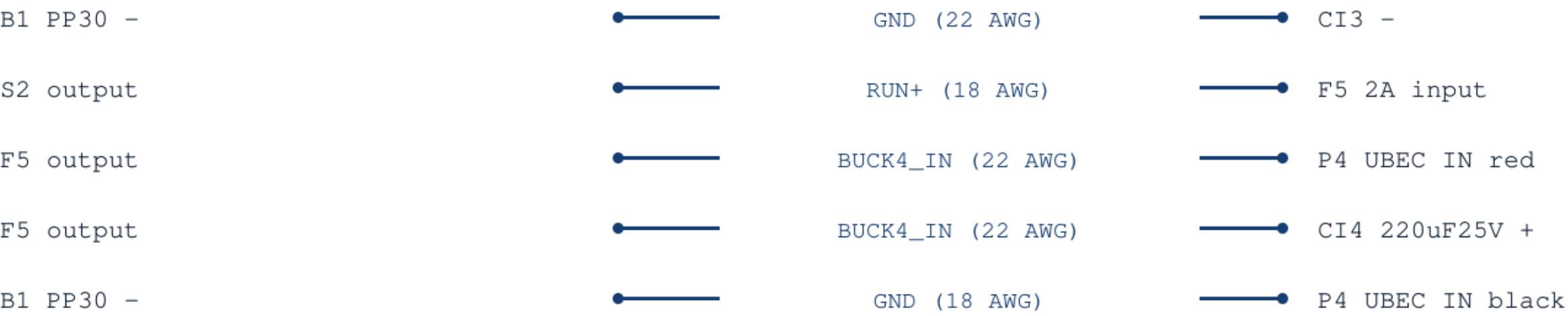
R2-24 circuit: power / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

B1 PP30 +		PACK+ (18 AWG)	F1 7.5A input
F1 output		FUSED+ (18 AWG)	S1 MAIN input
S1 output		MAIN+ (18 AWG)	S2 RUN input
S1 output		MAIN+ (22 AWG)	F6 1A input
F6 output		LOGIC_IN (22 AWG)	U6 MPM3610 VIN
S2 output		RUN+ (18 AWG)	F2 2A input
F2 output		BUCK1_IN (22 AWG)	P1 UBEC IN red
F2 output		BUCK1_IN (22 AWG)	CI1 220uF25V +
B1 PP30 -		GND (18 AWG)	P1 UBEC IN black
B1 PP30 -		GND (18 AWG)	P1 UBEC OUT black
B1 PP30 -		GND (22 AWG)	CI1 -
S2 output		RUN+ (18 AWG)	F3 2A input
F3 output		BUCK2_IN (22 AWG)	P2 UBEC IN red
F3 output		BUCK2_IN (22 AWG)	CI2 220uF25V +
B1 PP30 -		GND (18 AWG)	P2 UBEC IN black
B1 PP30 -		GND (18 AWG)	P2 UBEC OUT black
B1 PP30 -		GND (22 AWG)	CI2 -
S2 output		RUN+ (18 AWG)	F4 2A input
F4 output		BUCK3_IN (22 AWG)	P3 UBEC IN red
F4 output		BUCK3_IN (22 AWG)	CI3 220uF25V +
B1 PP30 -		GND (18 AWG)	P3 UBEC IN black
B1 PP30 -		GND (18 AWG)	P3 UBEC OUT black

R2-24 circuit: power / 2

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.



01-power-2.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 circuit: drive / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

P1 UBEC OUT red	●————	5V_LEFT (22 AWG)	————●	D1 DRV8833 VMotor +
P1 UBEC OUT red	●————	5V_LEFT (22 AWG)	————●	CO1 470uF10V +
U1 GPIO14	●————	LEFT_FWD (26 AWG)	————●	D1 AIN1
U1 GPIO14	●————	LEFT_FWD (26 AWG)	————●	D1 BIN1
U1 GPIO14	●————	LEFT_FWD (26 AWG)	————●	R_LEFT_FWD 10k pin1
U1 GPIO32	●————	LEFT_REV (26 AWG)	————●	D1 AIN2
U1 GPIO32	●————	LEFT_REV (26 AWG)	————●	D1 BIN2
U1 GPIO32	●————	LEFT_REV (26 AWG)	————●	R_LEFT_REV 10k pin1
U1 GPIO13	●————	ENABLE (26 AWG)	————●	D1 SLP
U1 GPIO36 / A4	●————	FAULT (26 AWG)	————●	D1 FLT
D1 AOUT1	●————	M1+ (22 AWG)	————●	M1 red
D1 AOUT2	●————	M1- (22 AWG)	————●	M1 black
M1 red	●————	M1+ (26 AWG)	————●	CM1 100nF pin1
M1 black	●————	M1- (26 AWG)	————●	CM1 pin2
D1 BOUT1	●————	M2+ (22 AWG)	————●	M2 red
D1 BOUT2	●————	M2- (22 AWG)	————●	M2 black
M2 red	●————	M2+ (26 AWG)	————●	CM2 100nF pin1
M2 black	●————	M2- (26 AWG)	————●	CM2 pin2
P2 UBEC OUT red	●————	5V_RIGHT (22 AWG)	————●	D2 DRV8833 VMotor +
P2 UBEC OUT red	●————	5V_RIGHT (22 AWG)	————●	CO2 470uF10V +
U1 GPIO15	●————	RIGHT_FWD (26 AWG)	————●	D2 AIN1
U1 GPIO15	●————	RIGHT_FWD (26 AWG)	————●	D2 BIN1

R2-24 circuit: drive / 2

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

U1 GPIO15		RIGHT_FWD (26 AWG)	R_RIGHT_FWD 10k pin1
U1 GPIO33		RIGHT_REV (26 AWG)	D2 AIN2
U1 GPIO33		RIGHT_REV (26 AWG)	D2 BIN2
U1 GPIO33		RIGHT_REV (26 AWG)	R_RIGHT_REV 10k pin1
U1 GPIO13		ENABLE (26 AWG)	D2 SLP
U1 GPIO36 / A4		FAULT (26 AWG)	D2 FLT
D2 AOUT1		M3+ (22 AWG)	M3 red
D2 AOUT2		M3- (22 AWG)	M3 black
M3 red		M3+ (26 AWG)	CM3 100nF pin1
M3 black		M3- (26 AWG)	CM3 pin2
D2 BOUT1		M4+ (22 AWG)	M4 red
D2 BOUT2		M4- (22 AWG)	M4 black
M4 red		M4+ (26 AWG)	CM4 100nF pin1
M4 black		M4- (26 AWG)	CM4 pin2
P3 UBEC OUT red		5V_REAR (22 AWG)	D3 DRV8833 VMotor +
P3 UBEC OUT red		5V_REAR (22 AWG)	CO3 470uF10V +
U1 GPIO27		REAR_FWD (26 AWG)	D3 AIN1
U1 GPIO27		REAR_FWD (26 AWG)	D3 BIN1
U1 GPIO27		REAR_FWD (26 AWG)	R_REAR_FWD 10k pin1
U1 GPIO12		REAR_REV (26 AWG)	D3 AIN2
U1 GPIO12		REAR_REV (26 AWG)	D3 BIN2
U1 GPIO12		REAR_REV (26 AWG)	R_REAR_REV 10k pin1

R2-24 circuit: drive / 3

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

U1 GPIO13	●————	ENABLE (26 AWG)	————●	D3 SLP
U1 GPIO36 / A4	●————	FAULT (26 AWG)	————●	D3 FLT
D3 AOUT1	●————	M5+ (22 AWG)	————●	M5 red
D3 AOUT2	●————	M5- (22 AWG)	————●	M5 black
M5 red	●————	M5+ (26 AWG)	————●	CM5 100nF pin1
M5 black	●————	M5- (26 AWG)	————●	CM5 pin2
D3 BOUT1	●————	M6+ (22 AWG)	————●	M6 red
D3 BOUT2	●————	M6- (22 AWG)	————●	M6 black
M6 red	●————	M6+ (26 AWG)	————●	CM6 100nF pin1
M6 black	●————	M6- (26 AWG)	————●	CM6 pin2
U1 GPIO13	●————	ENABLE (26 AWG)	————●	R_ENABLE 10k pin1
U1 GPIO36 / A4	●————	FAULT (26 AWG)	————●	R_FAULT 10k pin1
U1 GPIO13	●————	ENABLE (26 AWG)	————●	D4 SLP
U1 GPIO36 / A4	●————	FAULT (26 AWG)	————●	D4 FLT
D4 AOUT1	●————	M7+ (22 AWG)	————●	M7 red
M7 red	●————	M7+ (26 AWG)	————●	CM7 100nF pin1
D4 AOUT2	●————	M7- (22 AWG)	————●	M7 black
M7 black	●————	M7- (26 AWG)	————●	CM7 100nF pin2

02-drive-3.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 circuit: servo-and-sense / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

P4 UBEC OUT red	●	5V_SERVO (22 AWG)	●	SV1 steering red
P4 UBEC OUT red	●	5V_SERVO (26 AWG)	●	U7 AHCT125 pin14
P4 UBEC OUT red	●	5V_SERVO (22 AWG)	●	CS 1000uF10V +
P4 UBEC OUT red	●	5V_SERVO (26 AWG)	●	CB 100nF pin1
P4 UBEC OUT red	●	5V_SERVO (26 AWG)	●	R_POWER_TOP 10k pin1
B1 PP30 -	●	GND (22 AWG)	●	SV1 brown/black
B1 PP30 -	●	GND (26 AWG)	●	U7 pin7
B1 PP30 -	●	GND (26 AWG)	●	U7 pin1 / 10E
B1 PP30 -	●	GND (26 AWG)	●	U7 pin4 / 20E
B1 PP30 -	●	GND (26 AWG)	●	U7 pin10 / 30E
B1 PP30 -	●	GND (26 AWG)	●	U7 pin12 / 4A
B1 PP30 -	●	GND (26 AWG)	●	R_POWER_BOTTOM 15k pin2
P4 UBEC OUT red	●	5V_SERVO (26 AWG)	●	U7 pin13 / 40E
U1 GPIO25	●	STEER (26 AWG)	●	U7 pin2
U1 GPIO25	●	STEER (26 AWG)	●	R_STEER 10k pin1
U7 pin3	●	STEER_5V (26 AWG)	●	RS1 220ohm pin1
RS1 pin2	●	STEER_SIG (26 AWG)	●	SV1 signal
R_POWER_TOP pin2	●	RUN_SENSE (26 AWG)	●	U1 GPIO39 / A3
R_POWER_TOP pin2	●	RUN_SENSE (26 AWG)	●	R_POWER_BOTTOM pin1
R_POWER_TOP pin2	●	RUN_SENSE (26 AWG)	●	CP 100nF pin1
B1 PP30 -	●	GND (26 AWG)	●	CP pin2
S1 output	●	MAIN+ (26 AWG)	●	R_PACK_TOP 100k pin1

R2-24 circuit: servo-and-sense / 2

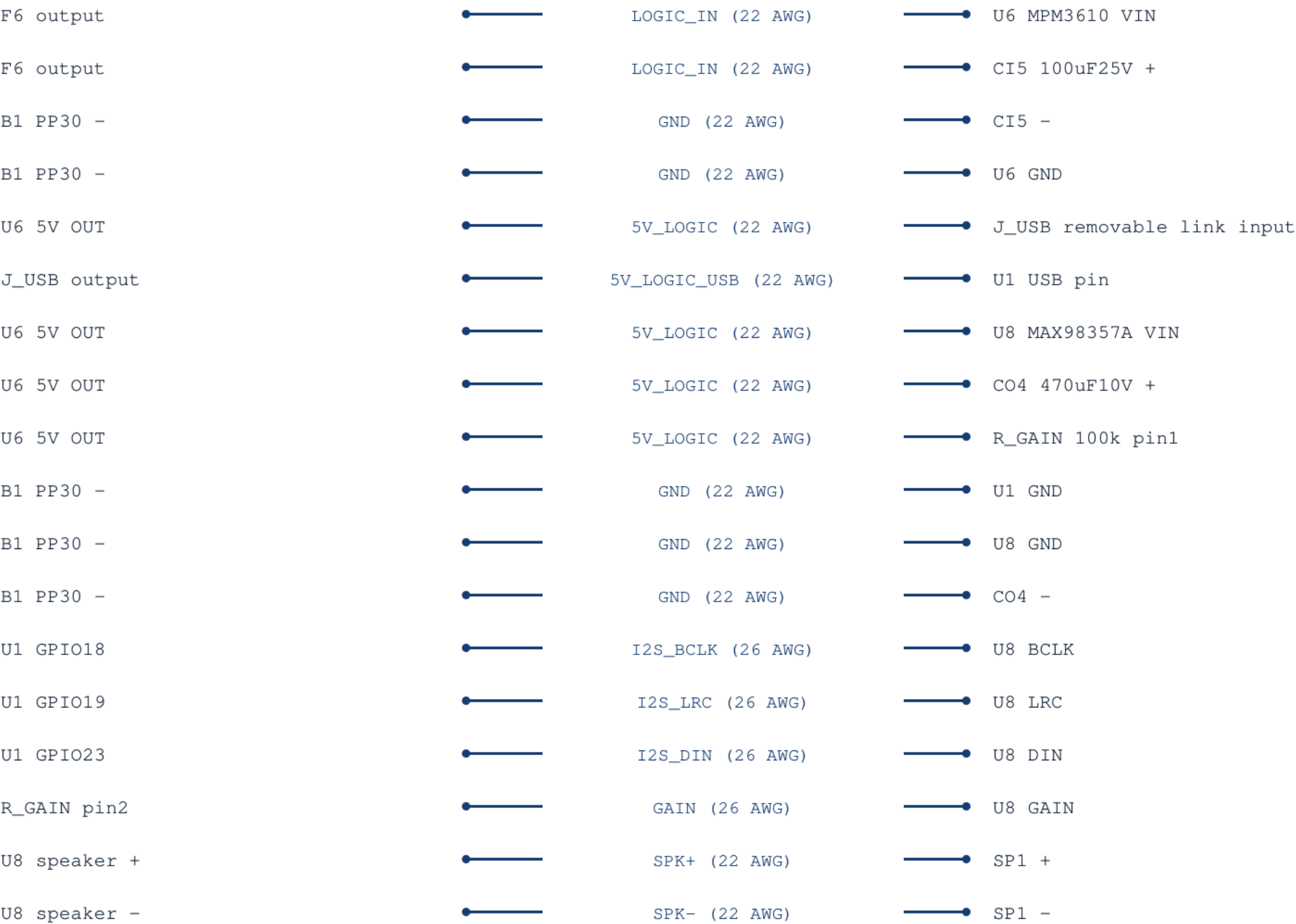
Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

R_PACK_TOP pin2	●————	PACK_ADC (26 AWG)	————●	U1 GPIO34 / A2
R_PACK_TOP pin2	●————	PACK_ADC (26 AWG)	————●	R_PACK_BOTTOM 22k pin1
R_PACK_TOP pin2	●————	PACK_ADC (26 AWG)	————●	CA 100nF pin1
B1 PP30 -	●————	GND (26 AWG)	————●	R_PACK_BOTTOM pin2
B1 PP30 -	●————	GND (26 AWG)	————●	CA pin2
F7 output	●————	HEAD_BUCK_IN (22 AWG)	————●	P5 UBEC IN red
F7 output	●————	HEAD_BUCK_IN (22 AWG)	————●	CI6 220uF25V +
P5 OUT red	●————	5V_HEAD (22 AWG)	————●	D4 VMotor +
P5 OUT red	●————	5V_HEAD (22 AWG)	————●	CO5 470uF10V +
U1 GPIO26	●————	HEAD_FWD (26 AWG)	————●	D4 AIN1
U1 GPIO26	●————	HEAD_FWD (26 AWG)	————●	R_HEAD_FWD 10k pin1
U1 GPIO17	●————	HEAD_REV (26 AWG)	————●	D4 AIN2
U1 GPIO17	●————	HEAD_REV (26 AWG)	————●	R_HEAD_REV 10k pin1
U1 GPIO4	●————	POST_EXTEND (26 AWG)	————●	U7 pin5
U1 GPIO16	●————	POST_RETRACT (26 AWG)	————●	U7 pin9
R_POT pin2	●————	POST_ADC (26 AWG)	————●	U9 AIN0
U9 AIN0	●————	POST_ADC (26 AWG)	————●	R_POT_FAIL 470k pin1
U9 AIN0	●————	POST_ADC (26 AWG)	————●	C_POT 100nF pin1

03-servo-and-sense-2.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 circuit: logic-and-audio / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.



04-logic-and-audio-1.svg | Named nets connect across sheets. Full return wiring is in the schedule.

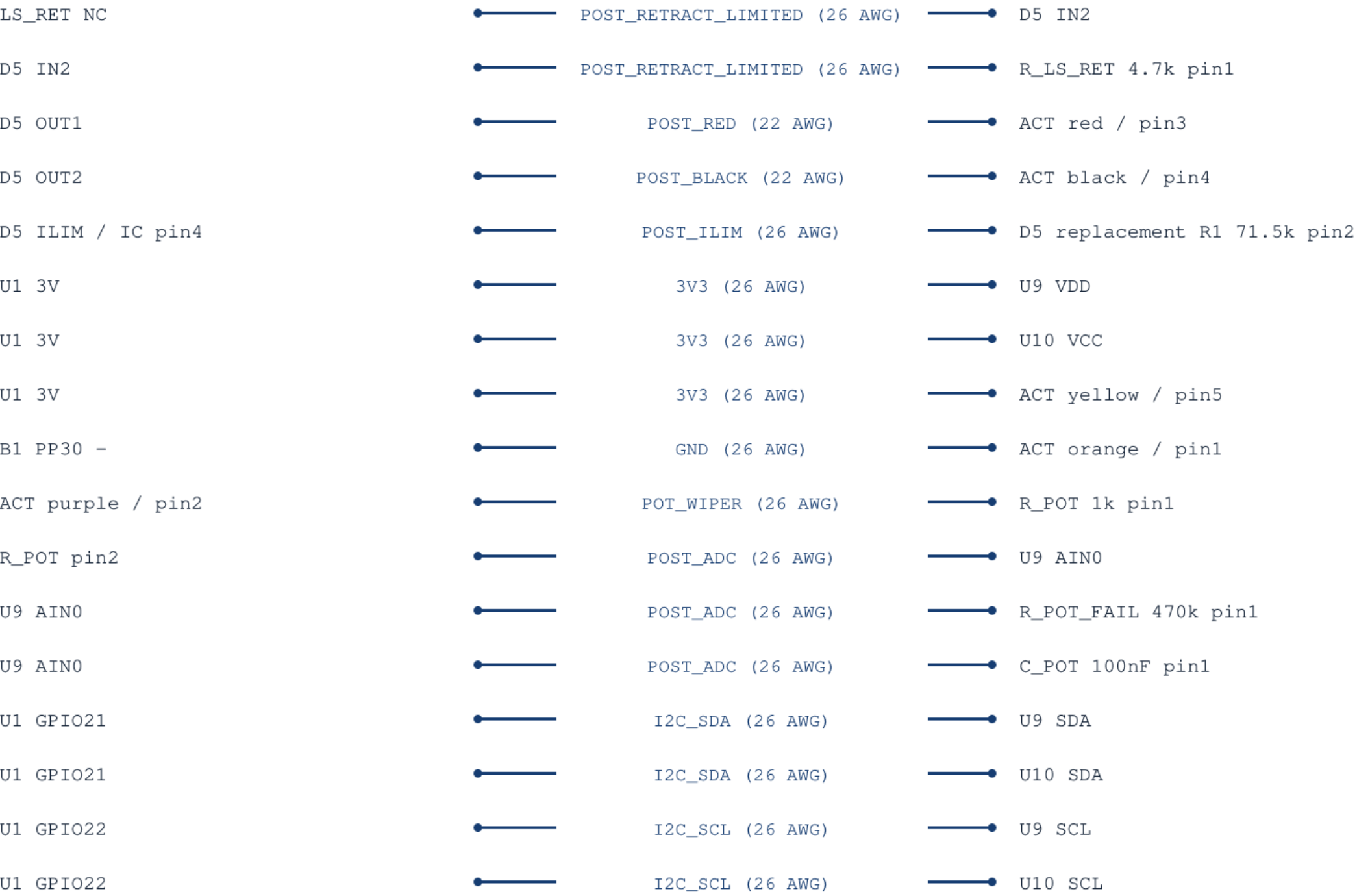
R2-24 circuit: posture / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

F8 output	●————	POST_BUCK_IN (22 AWG)	————●	P6 S13V25F12 VIN
F8 output	●————	POST_BUCK_IN (22 AWG)	————●	CI7 220uF25V +
P6 VOUT	●————	12V_POST_RAW (22 AWG)	————●	U10 INA219 VIN+
U10 VIN-	●————	12V_POST (22 AWG)	————●	D5 DRV8871 VM
U10 VIN-	●————	12V_POST (22 AWG)	————●	CO6 470uF25V +
B1 PP30 -	●————	GND (26 AWG)	————●	U10 GND
B1 PP30 -	●————	GND (26 AWG)	————●	U9 ADS1115 GND
B1 PP30 -	●————	GND (26 AWG)	————●	U9 ADDR
B1 PP30 -	●————	GND (26 AWG)	————●	U10 A0
B1 PP30 -	●————	GND (26 AWG)	————●	U10 A1
B1 PP30 -	●————	GND (26 AWG)	————●	U9 AIN1
B1 PP30 -	●————	GND (26 AWG)	————●	U9 AIN2
B1 PP30 -	●————	GND (26 AWG)	————●	U9 AIN3
U1 GPIO4	●————	POST_EXTEND (26 AWG)	————●	U7 pin5
U1 GPIO4	●————	POST_EXTEND (26 AWG)	————●	R_POST_EXTEND 10k pin1
U7 pin6	●————	POST_EXTEND_5V (26 AWG)	————●	LS_EXT COM
LS_EXT NC	●————	POST_EXTEND_LIMITED (26 AWG)	————●	D5 IN1
D5 IN1	●————	POST_EXTEND_LIMITED (26 AWG)	————●	R_LS_EXT 4.7k pin1
U1 GPIO16	●————	POST_RETRACT (26 AWG)	————●	U7 pin9
U1 GPIO16	●————	POST_RETRACT (26 AWG)	————●	R_POST_RETRACT 10k pin1
B1 PP30 -	●————	GND (26 AWG)	————●	R_POST_RETRACT pin2
U7 pin8	●————	POST_RETRACT_5V (26 AWG)	————●	LS_RET COM

R2-24 circuit: posture / 2

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.



05-posture-2.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 circuit: remaining-power-and-grounds / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

B1 PP30 -		GND (18 AWG)	P4 UBEC OUT black
B1 PP30 -		GND (22 AWG)	CI4 -
B1 PP30 -		GND (22 AWG)	D1 GND
B1 PP30 -		GND (22 AWG)	CO1 -
B1 PP30 -		GND (26 AWG)	R_LEFT_FWD pin2
B1 PP30 -		GND (26 AWG)	R_LEFT_REV pin2
B1 PP30 -		GND (22 AWG)	D2 GND
B1 PP30 -		GND (22 AWG)	CO2 -
B1 PP30 -		GND (26 AWG)	R_RIGHT_FWD pin2
B1 PP30 -		GND (26 AWG)	R_RIGHT_REV pin2
B1 PP30 -		GND (22 AWG)	D3 GND
B1 PP30 -		GND (22 AWG)	CO3 -
B1 PP30 -		GND (26 AWG)	R_REAR_FWD pin2
B1 PP30 -		GND (26 AWG)	R_REAR_REV pin2
B1 PP30 -		GND (26 AWG)	R_ENABLE pin2
U1 3V		3V3 (26 AWG)	R_FAULT pin2
B1 PP30 -		GND (26 AWG)	CS -
B1 PP30 -		GND (26 AWG)	CB pin2
B1 PP30 -		GND (26 AWG)	R_STEER pin2
S2 output		RUN+ (18 AWG)	F7 2A input
B1 PP30 -		GND (22 AWG)	P5 IN black
B1 PP30 -		GND (22 AWG)	P5 OUT black

R2-24 circuit: remaining-power-and-grounds / 2

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

B1 PP30 -	●————	GND (22 AWG)	————●	CI6 -
B1 PP30 -	●————	GND (22 AWG)	————●	CO5 -
B1 PP30 -	●————	GND (22 AWG)	————●	D4 GND
B1 PP30 -	●————	GND (22 AWG)	————●	D4 BIN1
B1 PP30 -	●————	GND (22 AWG)	————●	D4 BIN2
B1 PP30 -	●————	GND (26 AWG)	————●	R_HEAD_FWD pin2
B1 PP30 -	●————	GND (26 AWG)	————●	R_HEAD_REV pin2
S2 output	●————	RUN+ (18 AWG)	————●	F8 2A input
B1 PP30 -	●————	GND (22 AWG)	————●	P6 GND
B1 PP30 -	●————	GND (22 AWG)	————●	CI7 -
B1 PP30 -	●————	GND (22 AWG)	————●	CO6 -
B1 PP30 -	●————	GND (22 AWG)	————●	D5 GND
B1 PP30 -	●————	GND (26 AWG)	————●	R_POST_EXTEND pin2
B1 PP30 -	●————	GND (26 AWG)	————●	R_LS_EXT pin2
B1 PP30 -	●————	GND (26 AWG)	————●	R_LS_RET pin2
B1 PP30 -	●————	GND (26 AWG)	————●	D5 replacement R1 pin1
B1 PP30 -	●————	GND (26 AWG)	————●	R_POT_FAIL pin2
B1 PP30 -	●————	GND (26 AWG)	————●	C_POT pin2

06-remaining-power-and-grounds-2.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 | detailed assembled robot



REVISION C

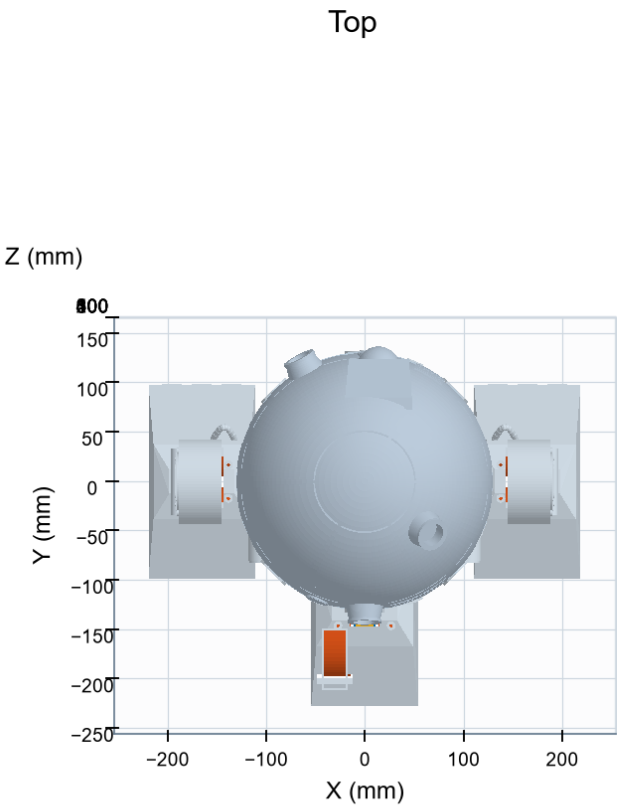
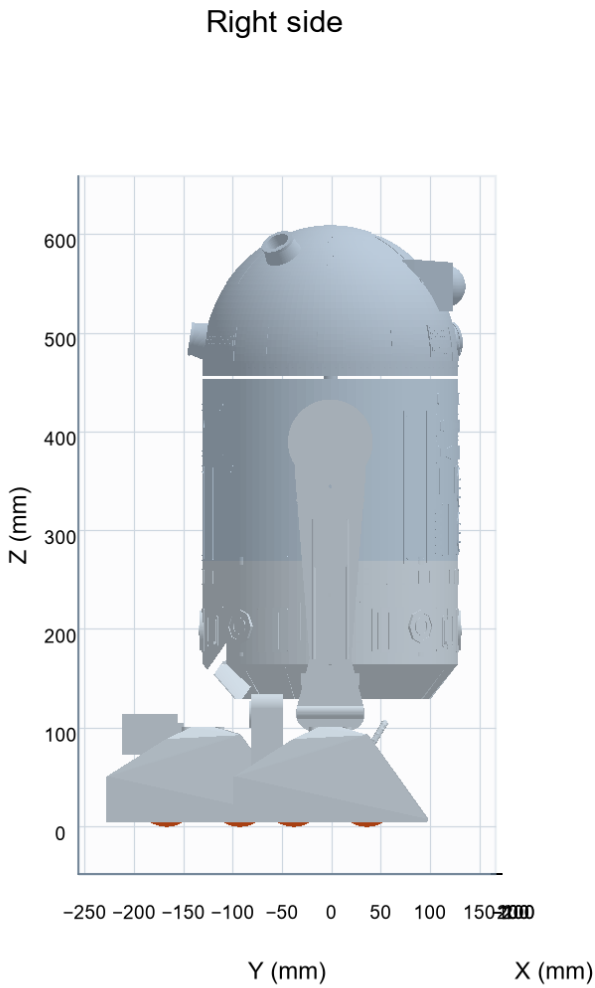
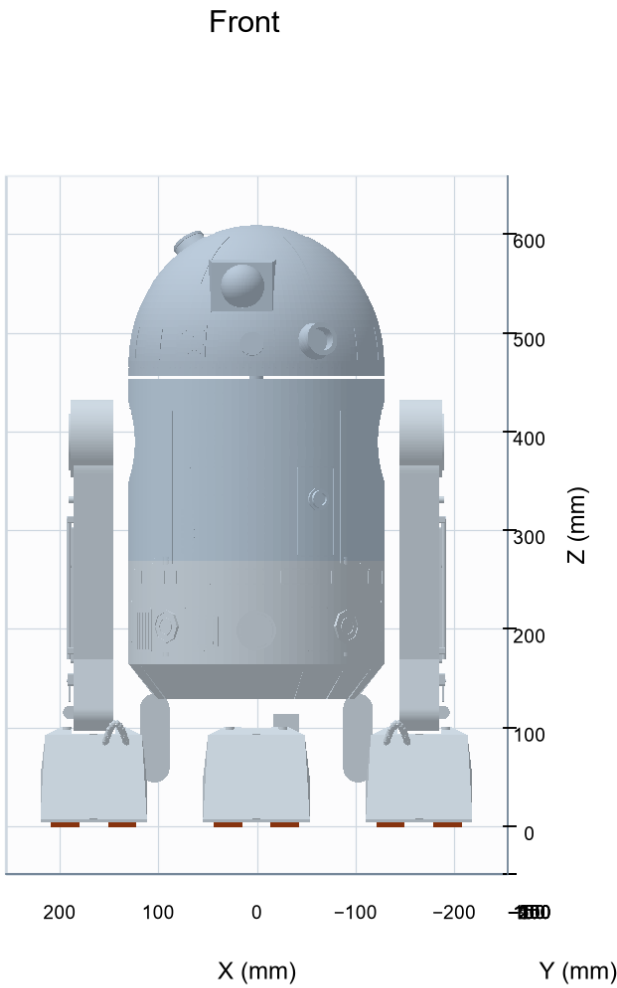
609.6 mm nominal height
259.25 mm body diameter
10 STL designs / 14 prints
Two whole body sections
One print per side leg

Metal shoulder pivots
Guided rear telescoping post
Supported body tilt
Internal head friction wheel
Six ground motors / 12 wheels
Phone control over Wi-Fi

Source CAD, with paint colors. Purchased parts are dimensional envelopes. Digital prototype; hardware untested.

01-robot-isometric.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | orthographic STL surfaces



Millimetres. Equal axis scale within each panel. Upright pose; exterior solids and wheel envelopes; paint omitted.

02-robot-orthographic.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | stackable covers separated

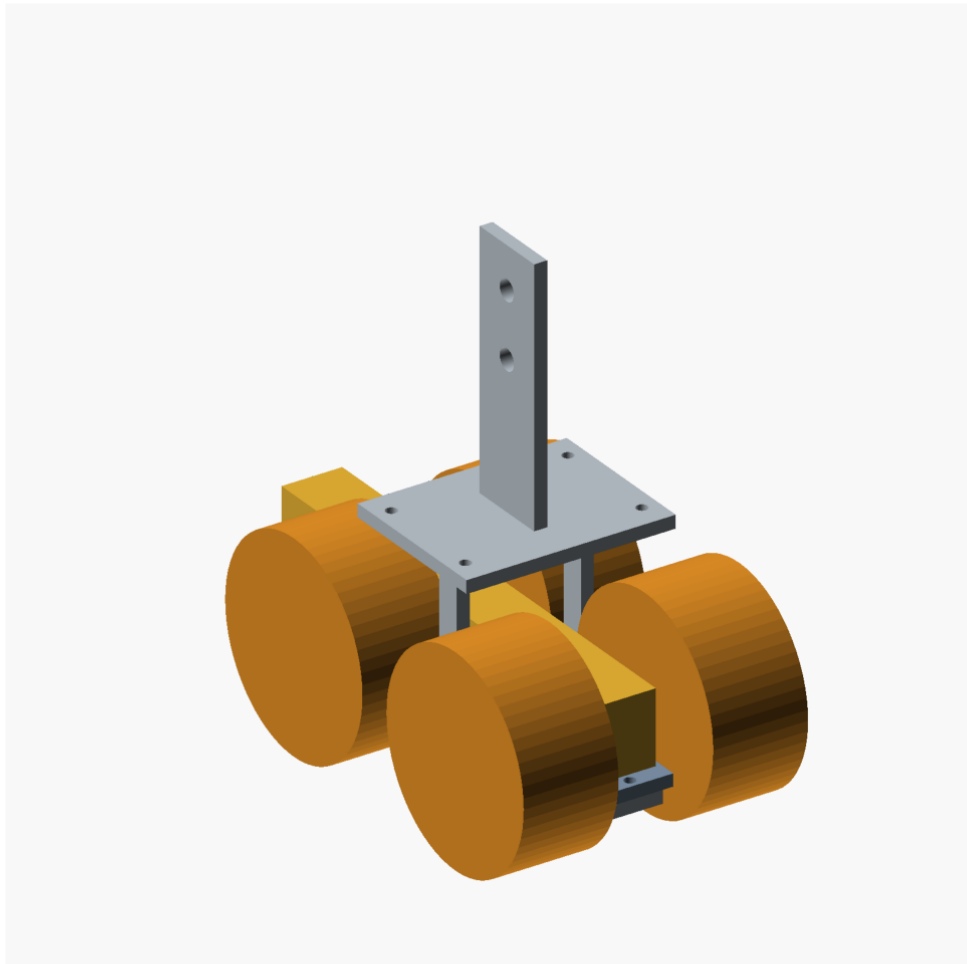


Assembly offsets are illustrative. Two full body rings, one dome, two side legs and three foot covers.

03-robot-exploded.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | internal mechanisms

Four-wheel foot | cover removed



Head drive | dome shell sectioned

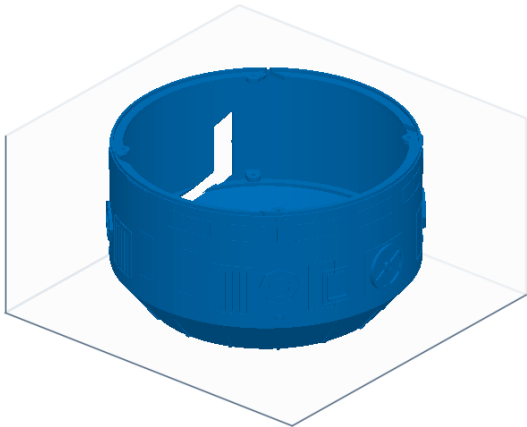


Actual mounting geometry; purchased motors/wheels/bearings are envelopes. Metal frame carries the vertical load.

04-mechanisms-exploded.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

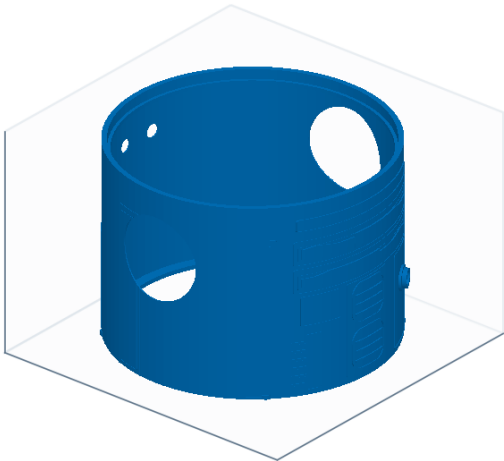
R2-24 | all ten printable components

body lower | qty 1



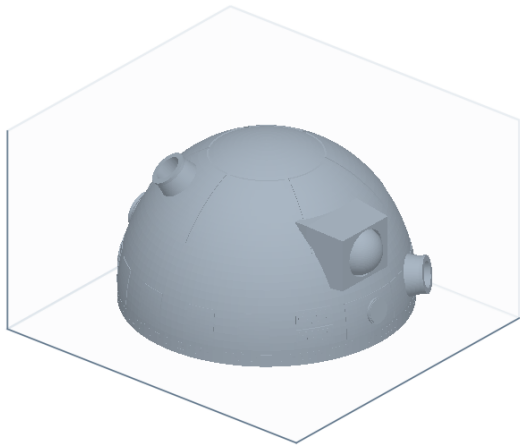
259.76 x 267.25 x 142.7 mm | PETG

body upper | qty 1



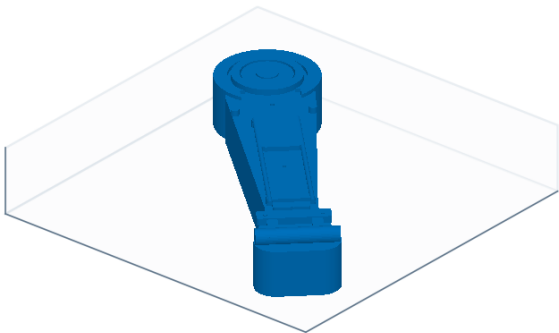
259.25 x 262.25 x 184.6 mm | PETG

dome | qty 1



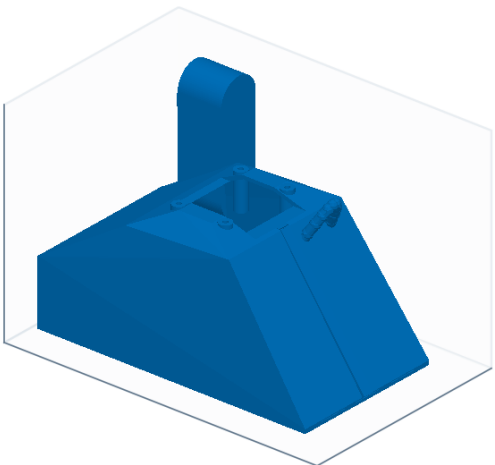
259.55 x 282.24 x 167.8 mm | PLA

leg | qty 2



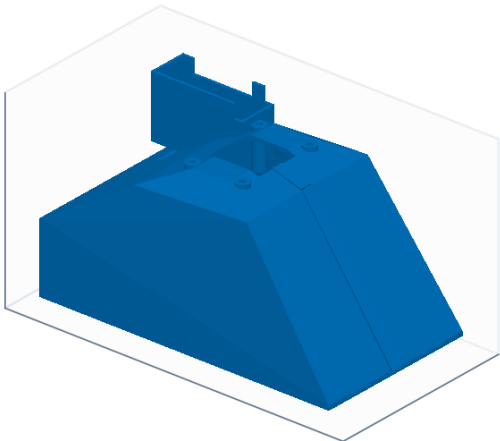
267.48 x 267.48 x 51 mm | PETG

outer foot | qty 2



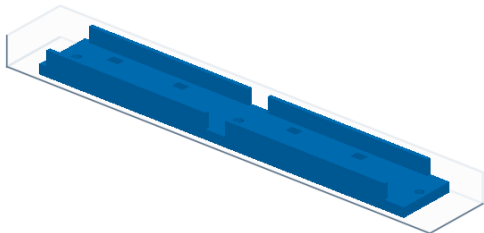
131 x 197.22 x 130 mm | PETG

rear foot | qty 1



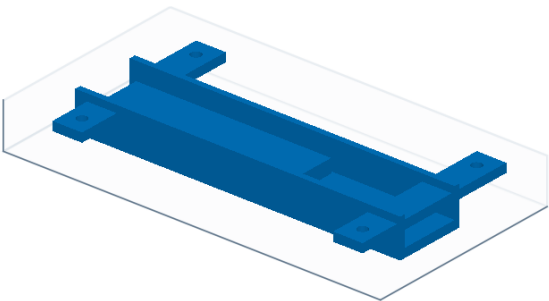
108 x 197.22 x 109 mm | PETG

drive cassette | qty 3



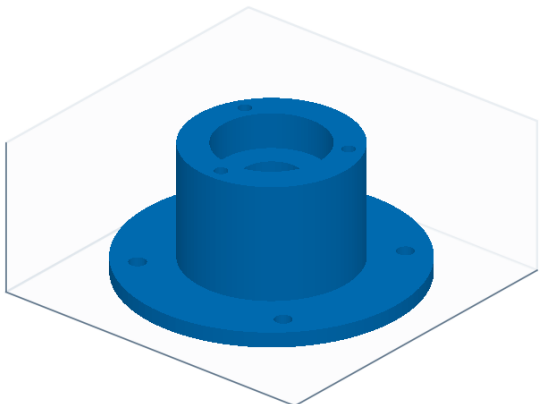
22.6 x 151 x 9.7 mm | PETG

head motor mount | qty 1



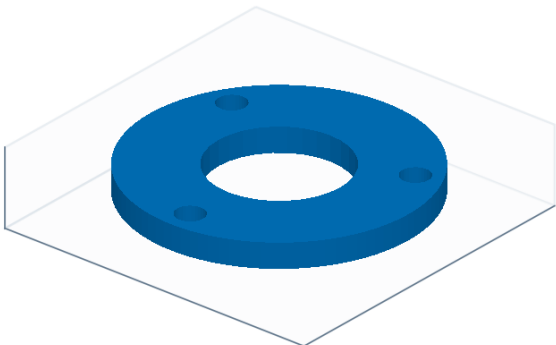
58 x 110 x 12.9 mm | PETG

bearing tower | qty 1



58 x 58 x 26 mm | PETG

bearing cap | qty 1



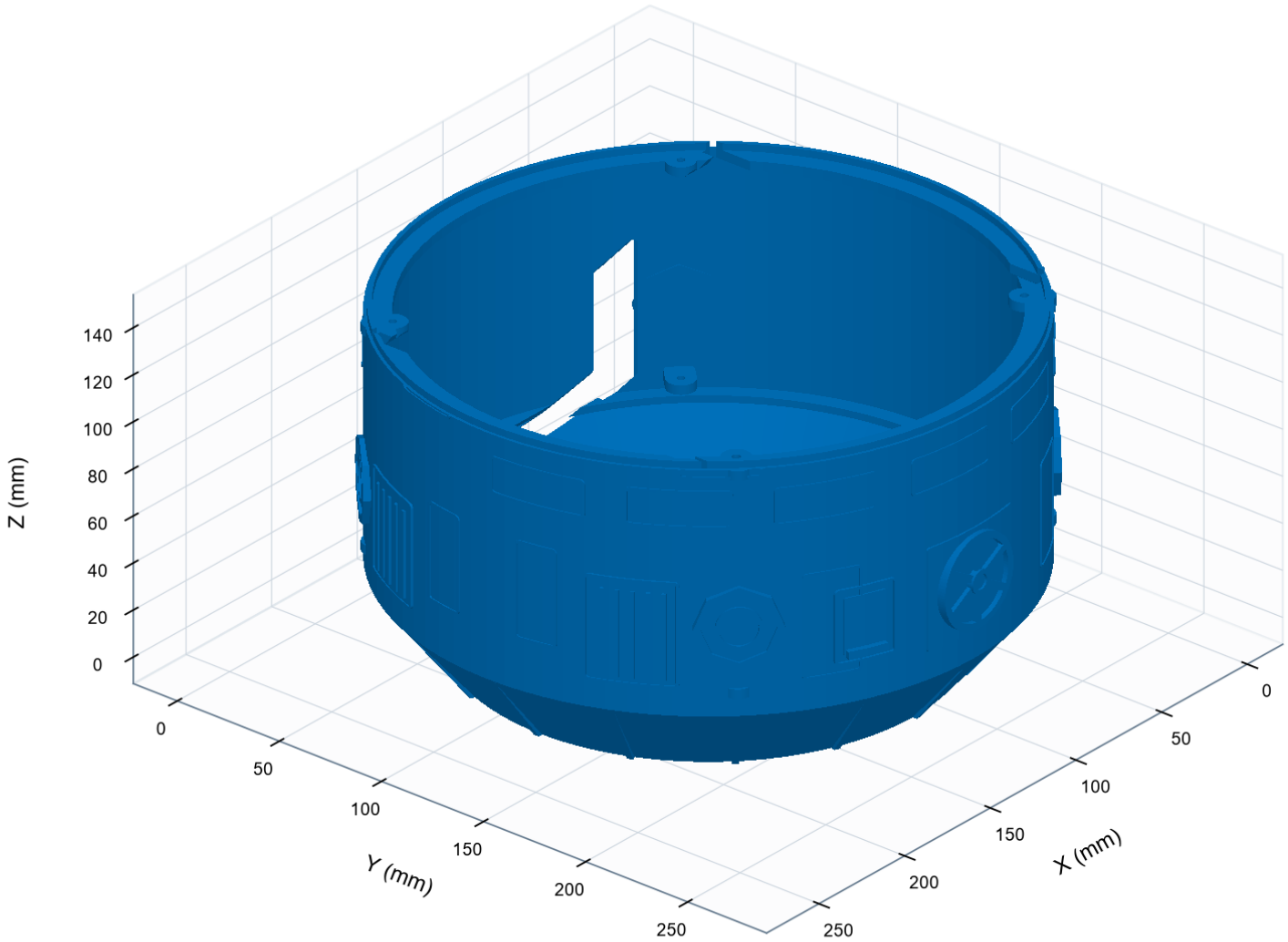
34 x 34 x 3 mm | PETG

Actual STL surfaces in their bed orientations; panels scaled independently. Print the outer foot once mirrored in X.

05-printed-components.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | body lower

body lower | qty 1



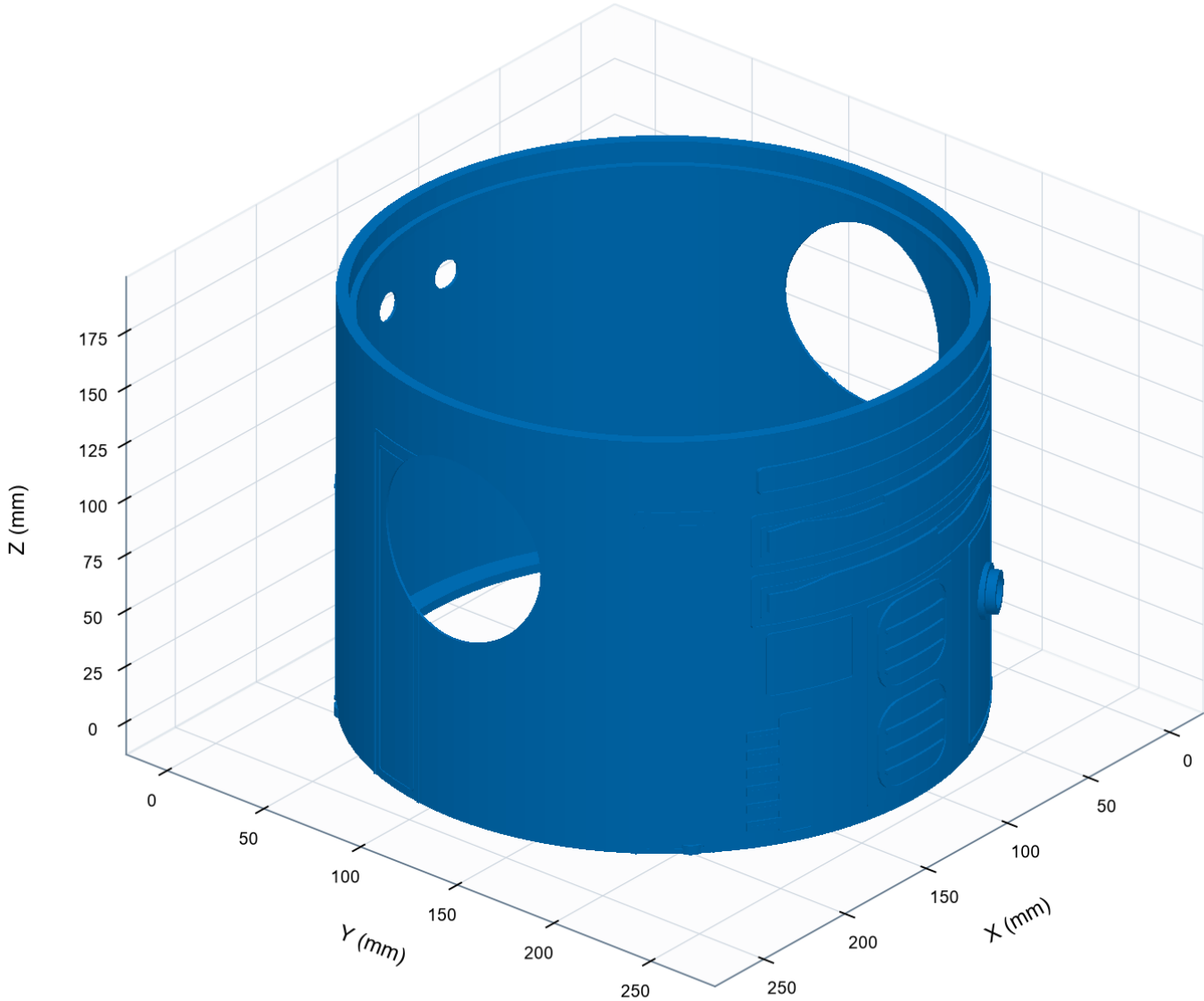
259.76 x 267.25 x 142.7 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/01-body_lower.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | body upper

body upper | qty 1



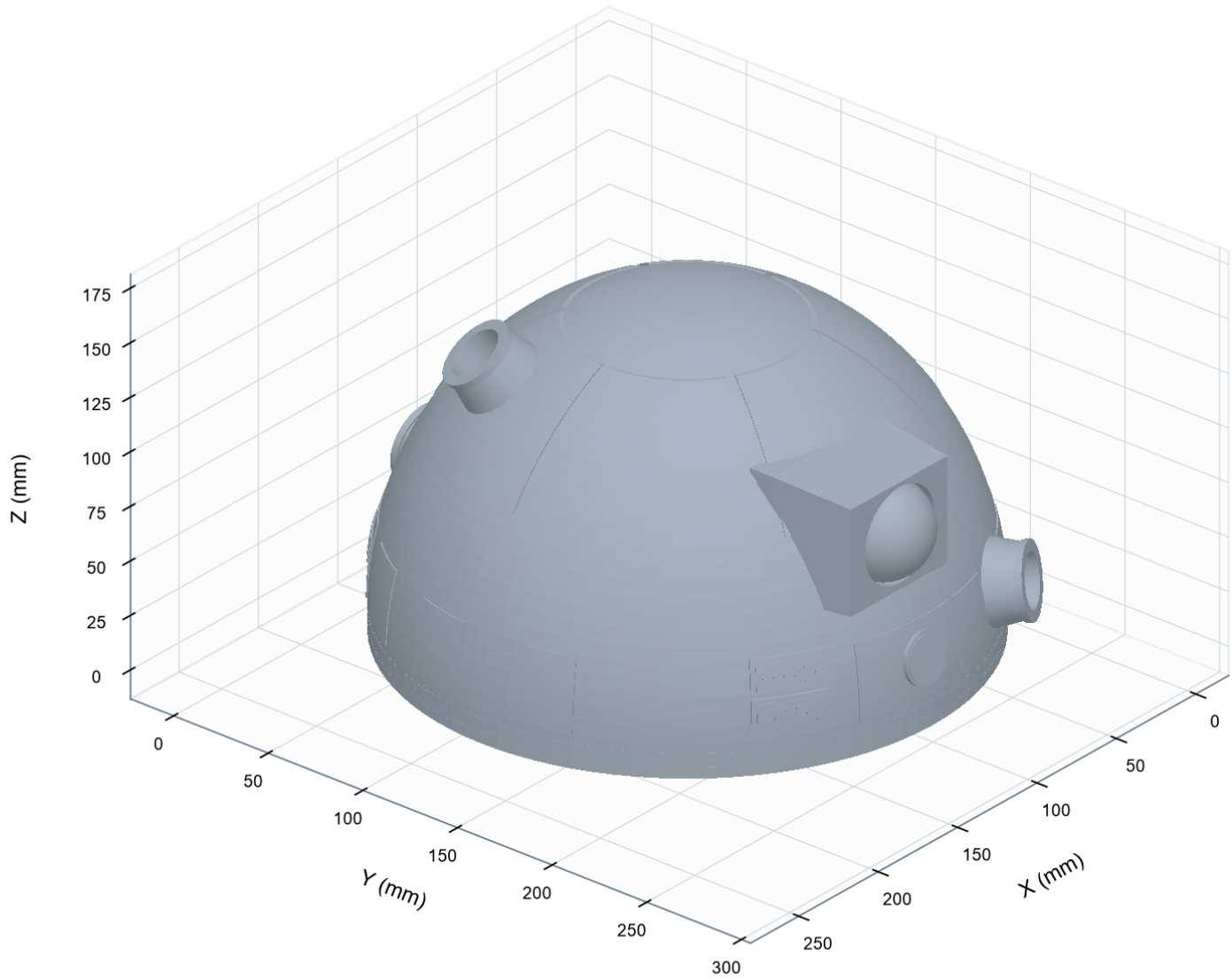
259.25 x 262.25 x 184.6 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/02-body_upper.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | dome

dome | qty 1



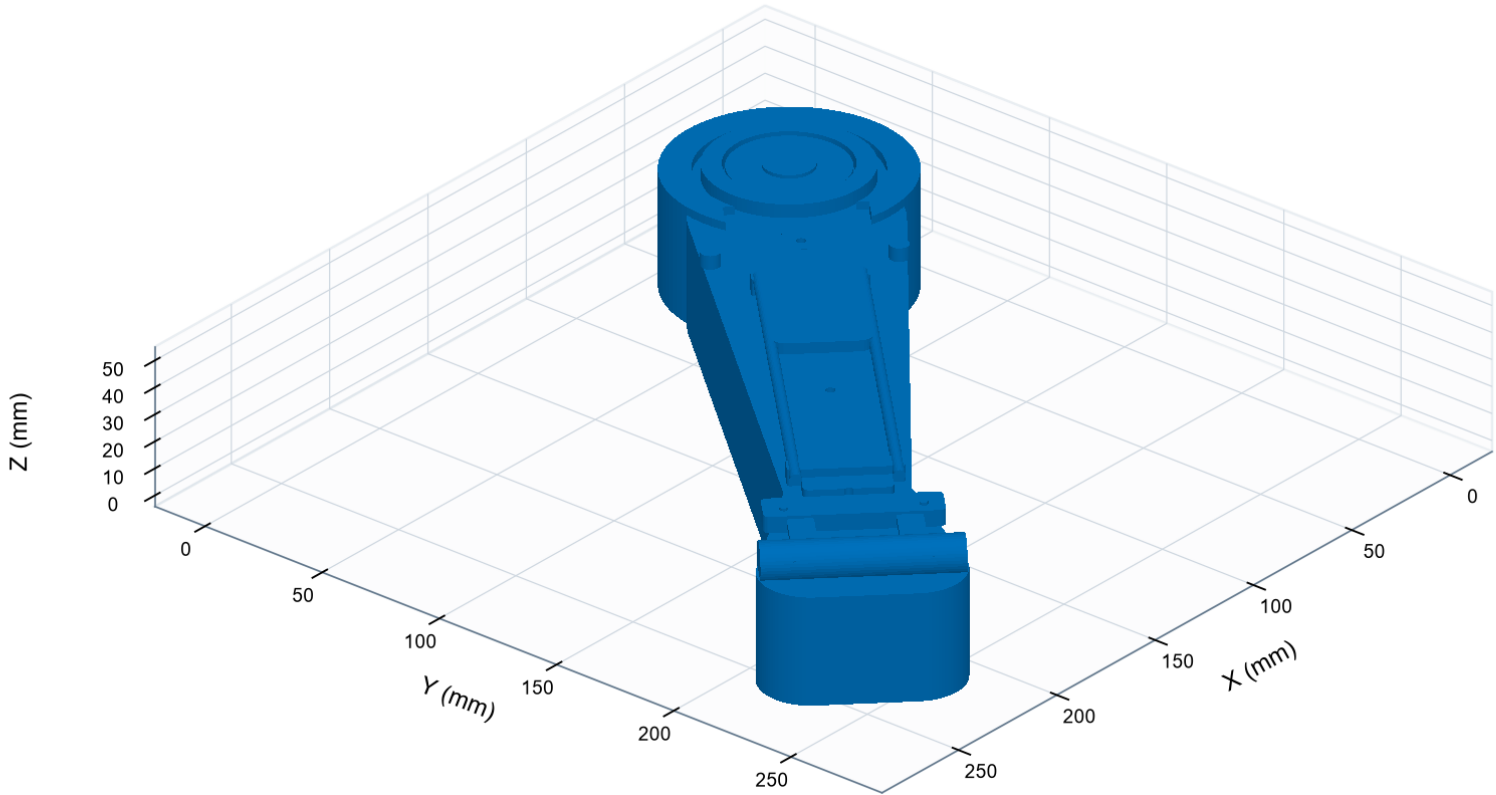
259.55 x 282.24 x 167.8 mm | PLA

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/03-dome.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | leg

leg | qty 2



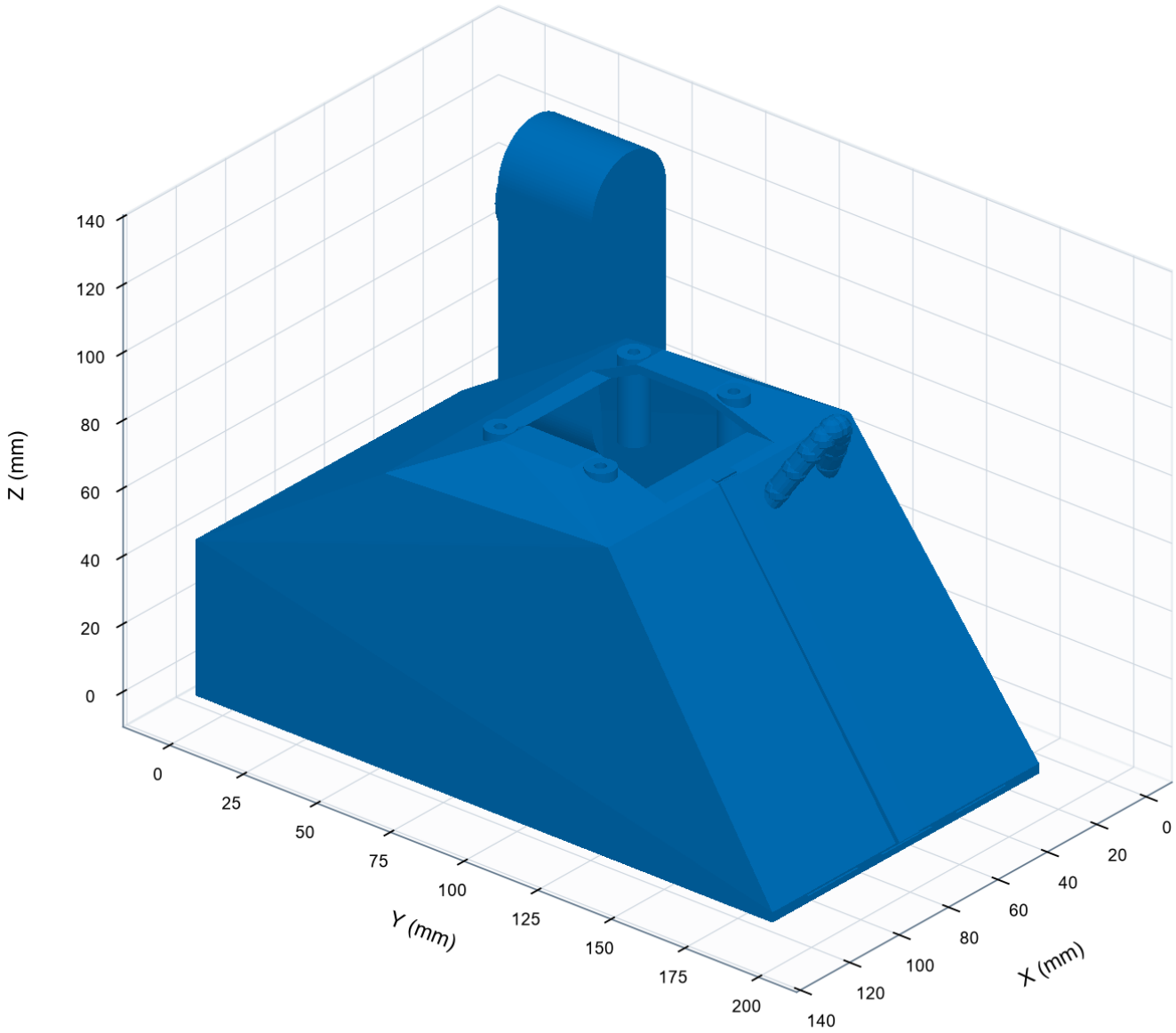
267.48 x 267.48 x 51 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/04-leg.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | outer foot

outer foot | qty 2



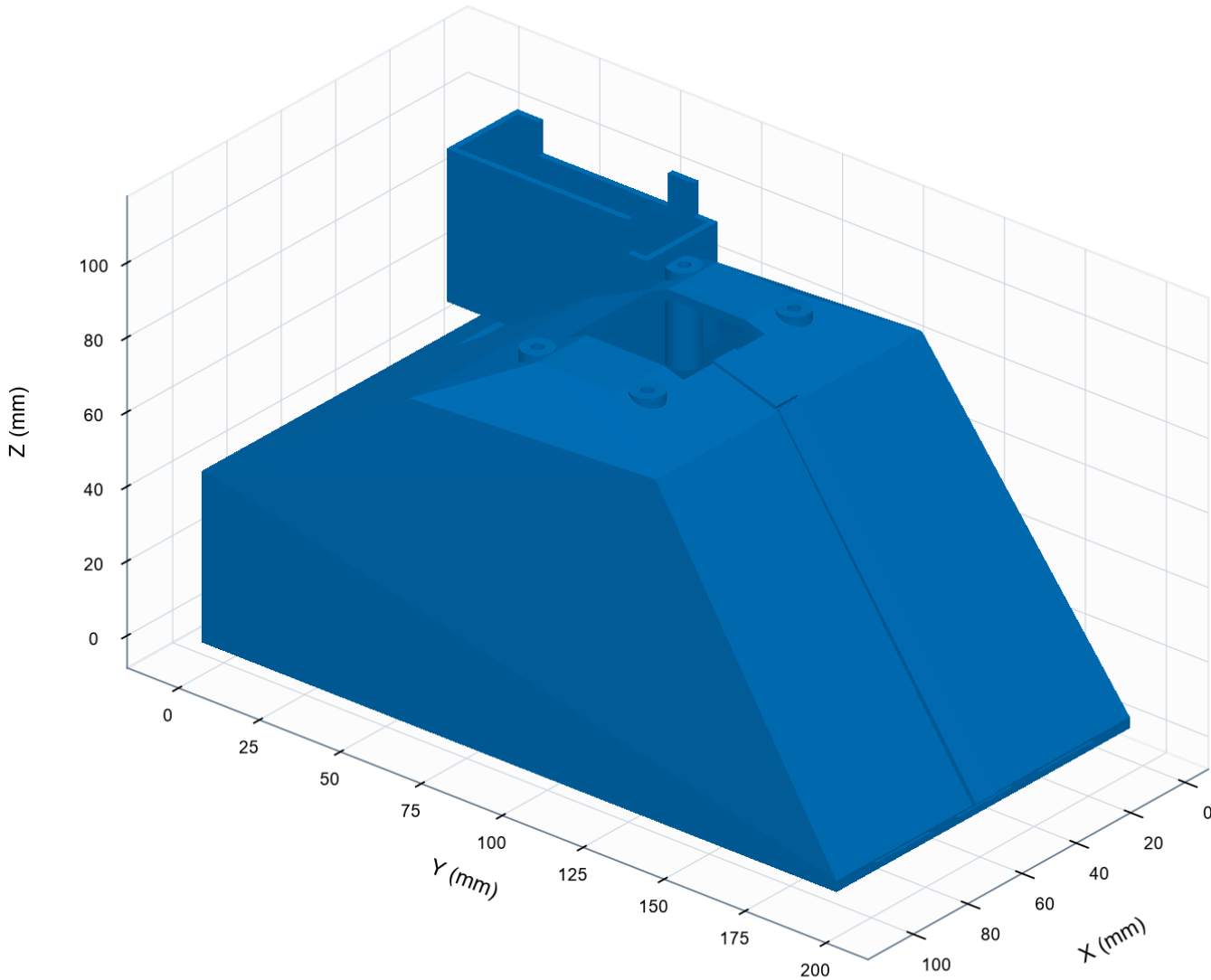
131 x 197.22 x 130 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/05-outer_foot.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | rear foot

rear foot | qty 1



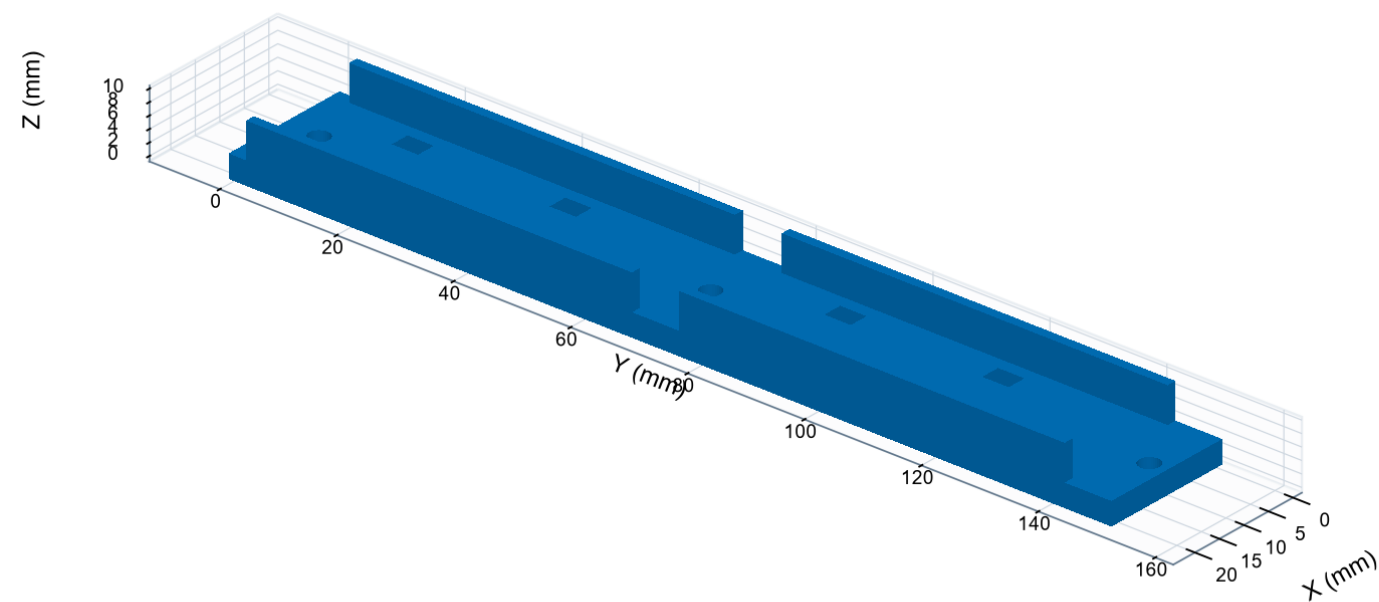
108 x 197.22 x 109 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/06-rear_foot.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | drive cassette

drive cassette | qty 3



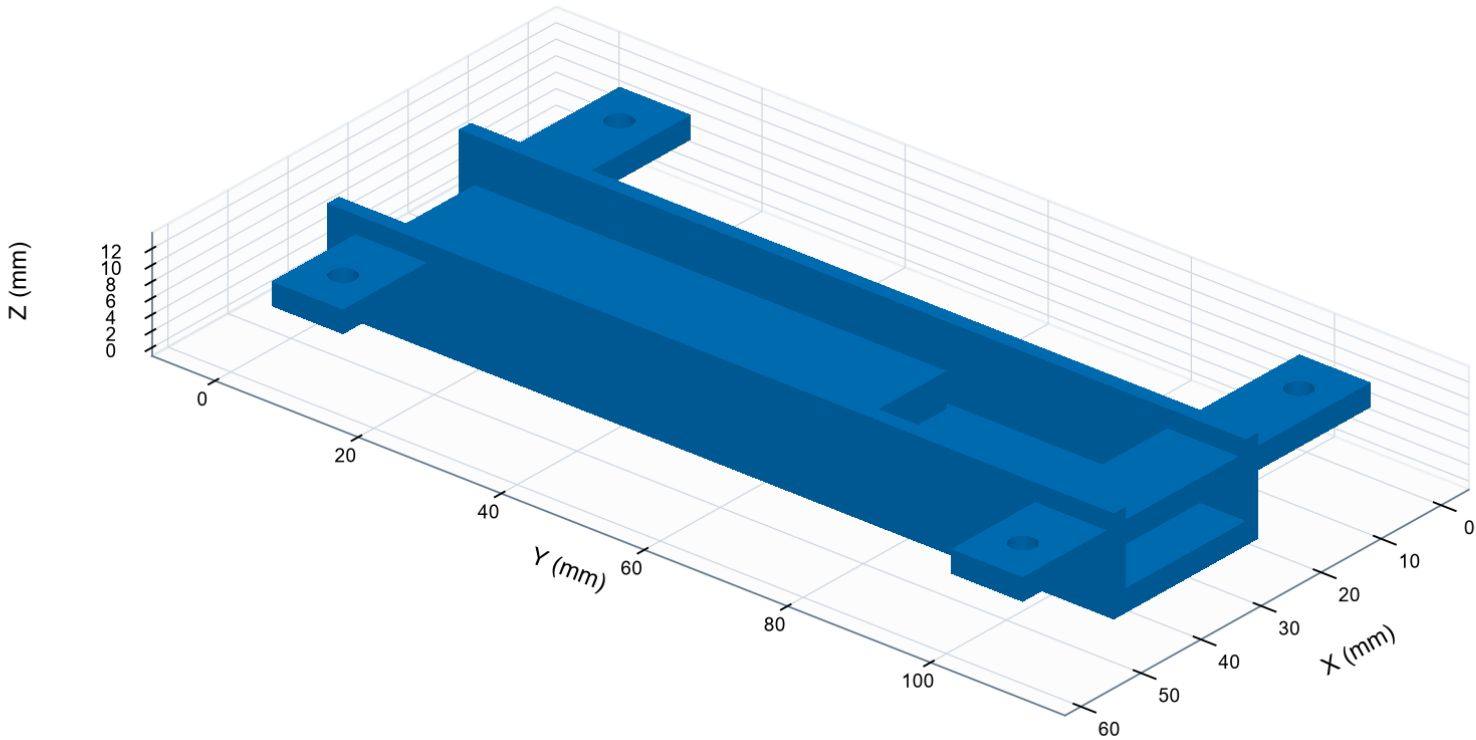
22.6 x 151 x 9.7 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/07-drive_cassette.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | head motor mount

head motor mount | qty 1



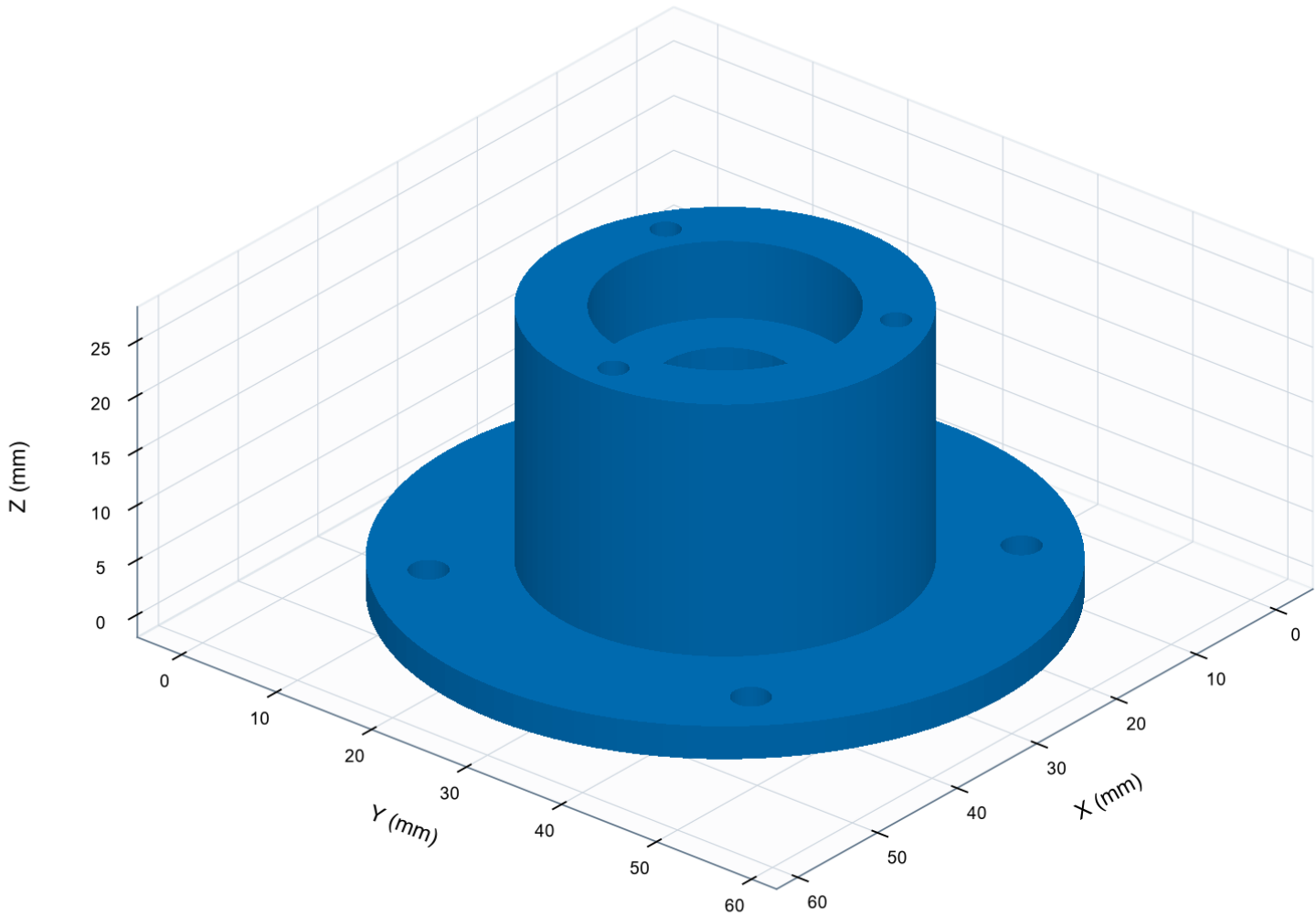
58 x 110 x 12.9 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/08-head_motor_mount.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | bearing tower

bearing tower | qty 1



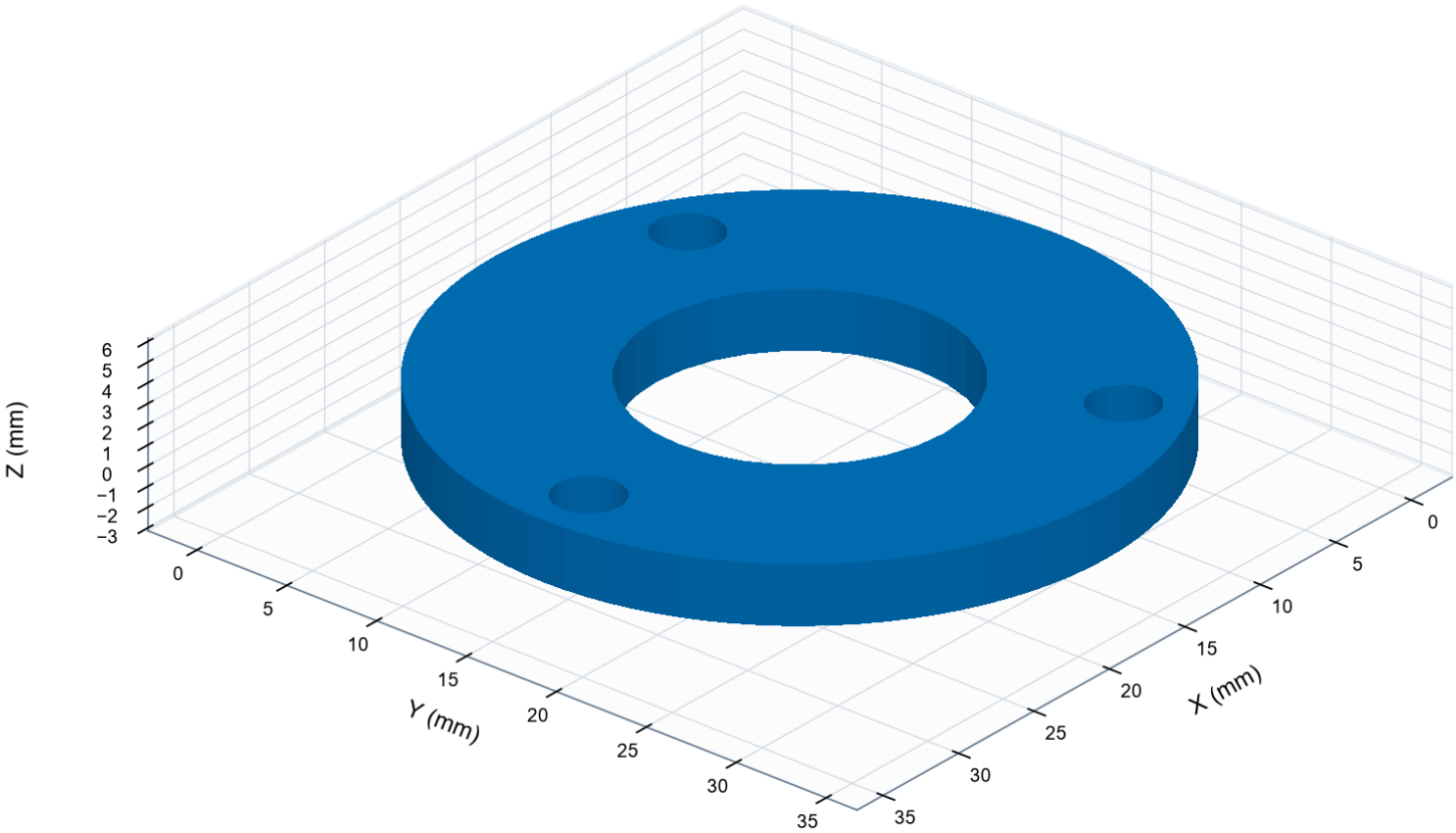
58 x 58 x 26 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/09-bearing_tower.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | bearing cap

bearing cap | qty 1

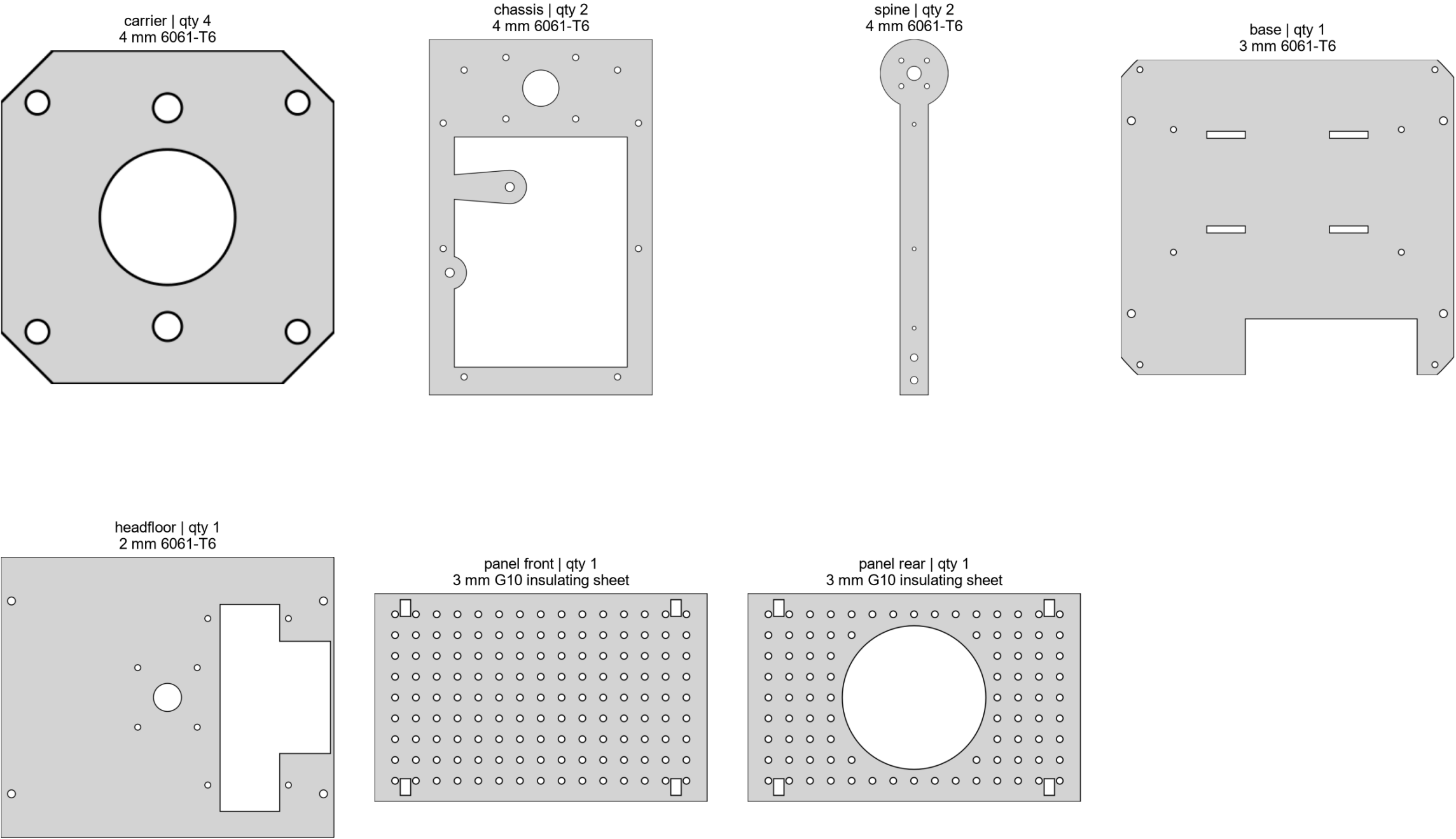


34 x 34 x 3 mm | PETG

Closed single-solid STL. Native bed orientation; dimensions in millimetres. See the print manifest for quantity.

components/10-bearing_cap.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

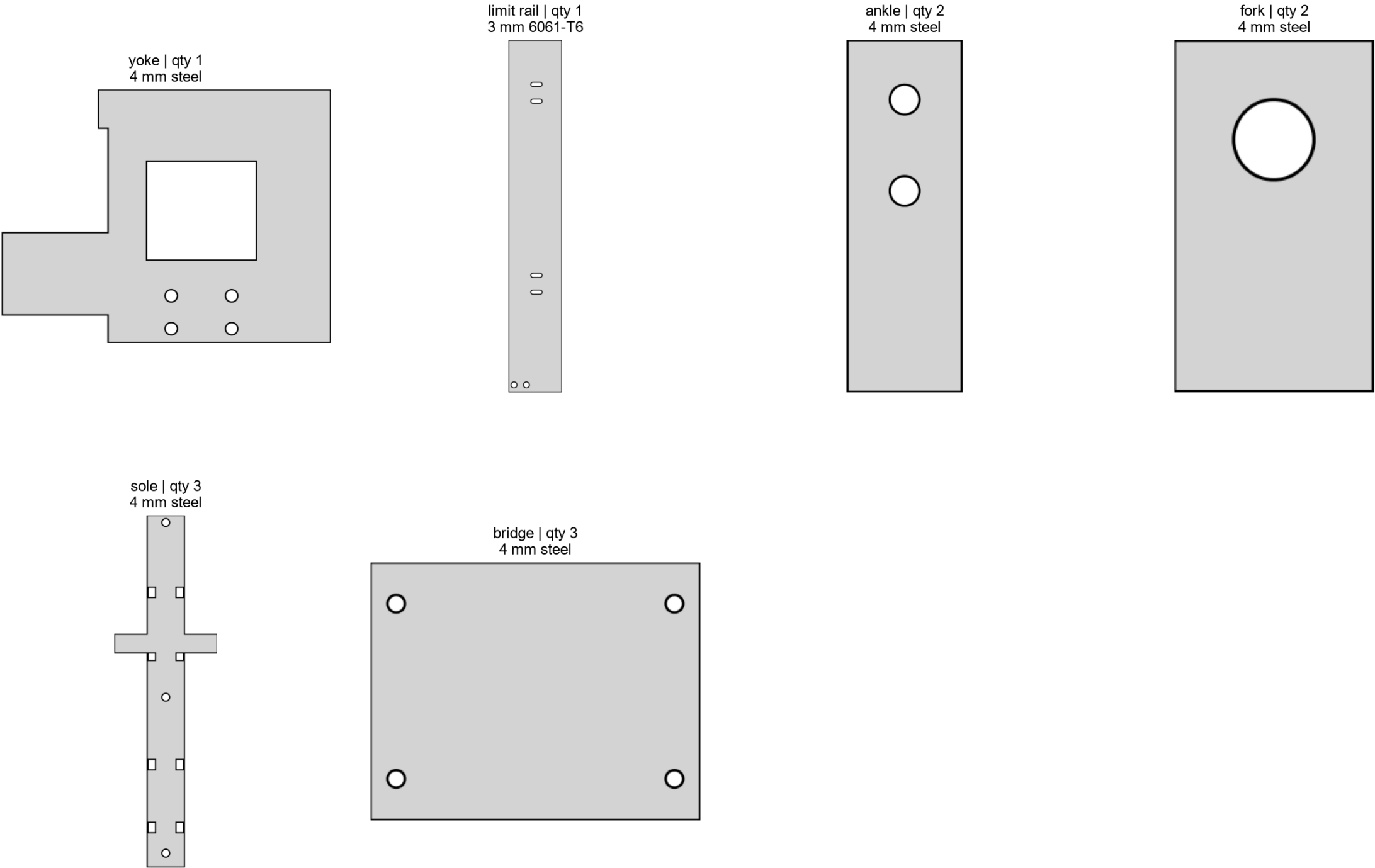
R2-24 | metal and insulating-sheet cut profiles 1



Exact-size DXF/SVG profiles are in cad/metal. Images are not paper templates. Read fabrication.md for bores, threads and welds.

06-metal-profiles.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

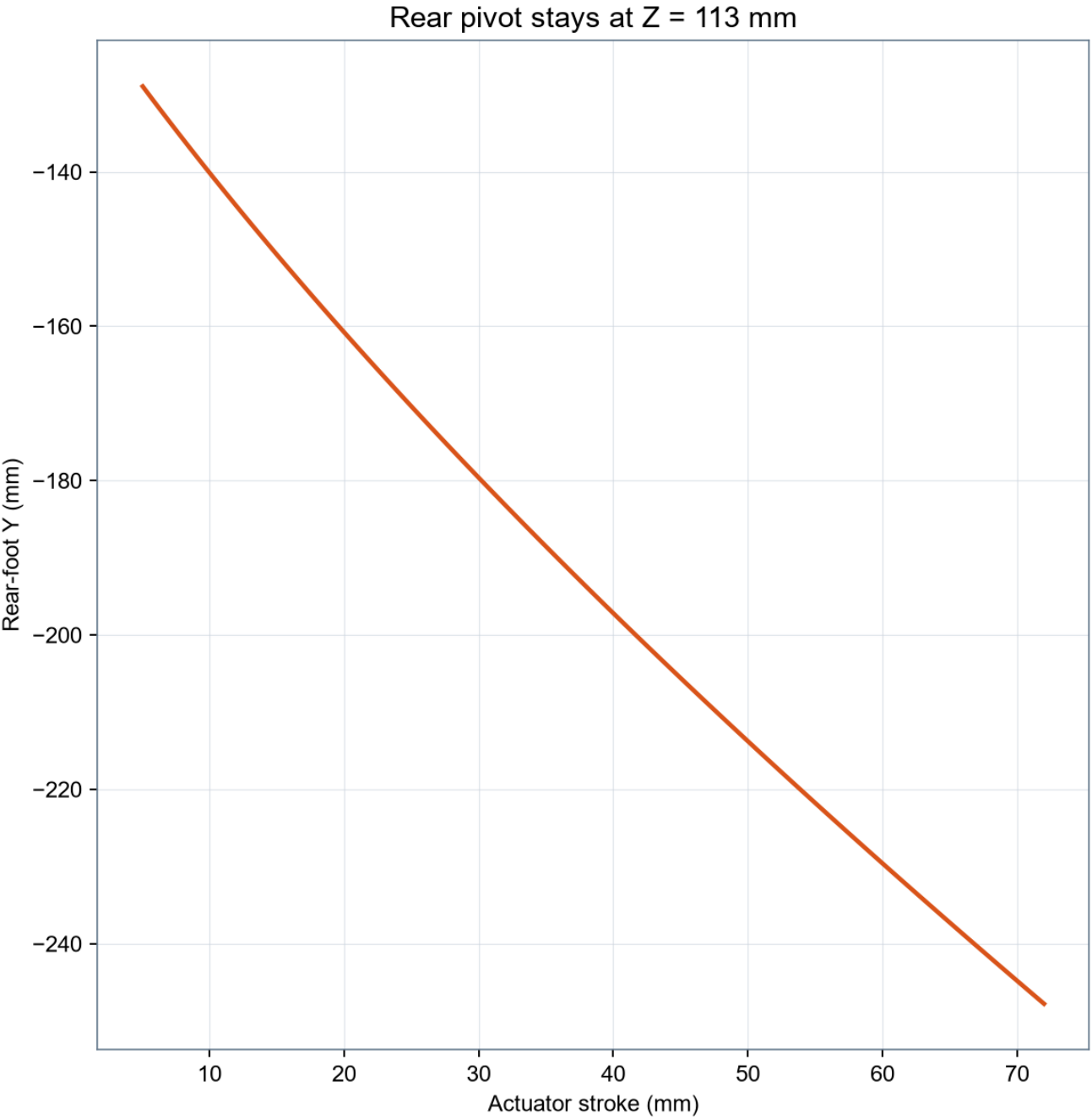
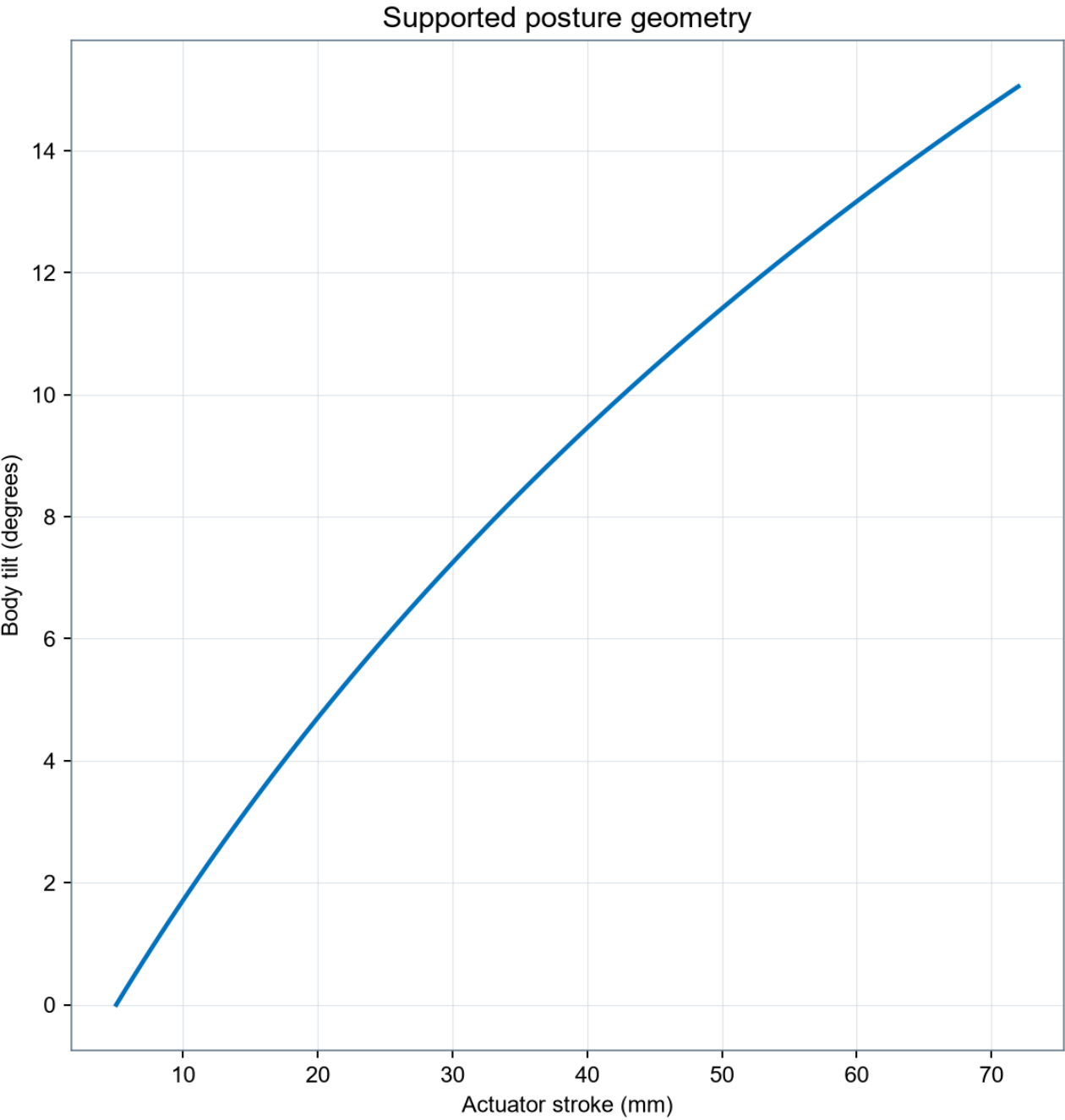
R2-24 | metal and insulating-sheet cut profiles 2



Exact-size DXF/SVG profiles are in cad/metal. Images are not paper templates. Read fabrication.md for bores, threads and welds.

07-metal-profiles.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.

R2-24 | rear post and body tilt



Calculated kinematics only. Rear placement follows the requested layout; all three feet stay down. Not measured hardware motion.

08-posture-geometry.png | Original PNG is included in the drawing collection. Cut-profile images are not paper templates.